ESE 4025 Numericals: interactive solutions

⌂ All materials

HVAC&R System Design · past papers 2020, 2021, 2023, 2024

How to read this pack. Every numerical has the same parts: **A** problem, **B** how it works, **C** interactive graph, **D** the exact table rows used, **E** the whole P–h chart with the cycle drawn, **F** the model answer, **G** common mistakes. Grey boxes UNDERSTAND are for your understanding only. Green boxes WRITE IN EXAM show exactly what to write in the answer script, and how much.

Contents

1. 1. R22 cycle, 80 kW (2024 1c, 2021 1c)
2. 2. R22 standard cycle, 50 kW (2020 2c)
3. 3. Ammonia ice plant (2023 2c)
4. 4. Ammonia ice plant (2021 2c)
5. 5. Ton of refrigeration loads (2024 1a, 2023 1d, 2020 1c)
6. 6. Aircraft air cycle (2020 3c)
7. 7. Summer AC with bypass factor (2024 5c)
8. 8. ADP from room loads (2021 5c)
9. 9. CO_2 ventilation, Dhaka vs Khulna (2024 6c)
10. 10. Room cooling load (2024 7c)
11. 11. Duct sizing, round (2020 6c)
12. 12. Duct sizing, rectangular 3:1 (2021 8d)
13. 13. Duct sizing (2023 8d)
14. 14. Duct sizing with fitting losses (2024 8c)

Chart library · Abbreviations

1. R22 VCR cycle with superheat and subcooling: 2024 1(c) and 2021 1(c)

Topic A4 · 13 + 15 marks ·

A. Problem

A refrigeration system using R22 is to have a refrigerating capacity of 80 kW. The system follows a VCR (vapour compression refrigeration) cycle with 5 °C subcooling and 10 °C superheating, in which the evaporating and condensing temperatures are −10 °C and 45 °C. Determine: (i) the volume flow rate of refrigerant, (ii) COP, (iii) the power required to drive the compressor. *(Same question word for word in 2021 1(c), 15 marks.)*

B. UNDERSTAND How the cycle/process works

For understanding – do not write

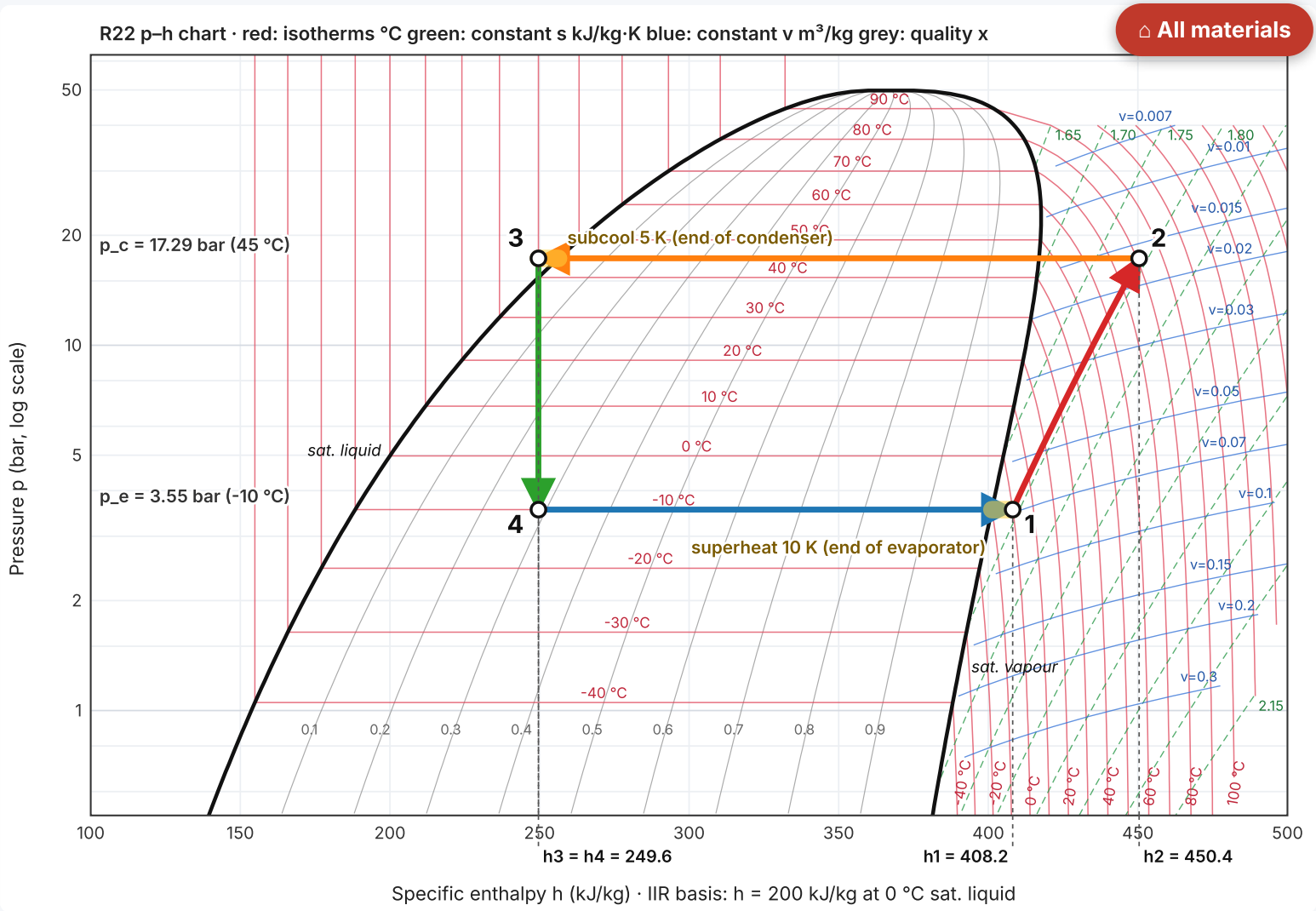
Where things happen. Superheating happens at the **end of the evaporator**: the refrigerant finishes boiling at −10 °C, then warms 10 K more, to 0 °C, while still at evaporator pressure. **Subcooling** happens at the **end of the condenser**: the vapour finishes condensing at 45 °C, then the liquid cools 5 K more, to 40 °C, while still at condenser pressure.

Point	Where	State	Found on the P–h chart as the crossing of
1	leaving evaporator, entering compressor	superheated vapour, 0 °C	horizontal line p_e (sat. pressure at −10 °C) × isotherm 0 °C (right of dome)
2	leaving compressor	superheated vapour, about 81 °C	horizontal line p_c (sat. pressure at 45 °C) × constant-s line through 1
3	leaving condenser	subcooled liquid, 40 °C	horizontal line p_c × isotherm 40 °C (vertical, left of dome) ≈ h_f at 40 °C
4	leaving expansion valve	wet mixture at −10 °C	vertical line down from 3 ($h_4 = h_3$) × horizontal line p_e

Why $h_3 = h_f$(40 °C)? Liquid enthalpy hardly depends on pressure, so subcooled liquid at 40 °C has (almost) the same h as saturated liquid at 40 °C. That is the standard textbook step.

What you need: h_1, h_2, h_3 (= h_4) and v_1. Everything else is arithmetic: RE (refrigerating effect) = $h_1 - h_4$, $w = h_2 - h_1$, $m = Q/RE$, $V = m\, v_1$.

C. UNDERSTAND Interactive graph



Hover over the chart to read h, p, t, s and v at any point. Press **Step through** to follow the refrigerant round the cycle.

p_e	**3.548**	bar	p_c	**17.292**	bar	h_1	**408.2**	kJ/kg
h_2	**450.4**	kJ/kg	$h_3 = h_4$	**249.6**	kJ/kg	v_1	**0.0686**	m³/kg
t_2 (discharge)	**80.8**	°C	x_4		**0.288**	RE = $h_1 - h_4$	**158.5**	kJ/kg
$w = h_2 - h_1$	**42.3**	kJ/kg	$\dot{m} = Q/RE$	**0.5047**	kg/s	$\dot{V} = \dot{m}\, v_1$	**0.0346**	m³/s
COP = RE/w		**3.75**	Power = $\dot{m}\, w$	**21.33**	kW	Q_{cond}	**101.3**	kW
p_c/p_e		**4.87**						

Working (formula = substitution = answer)

RE = $h_1 - h_4$ = 408.2 − 249.6 = **158.5 kJ/kg**

$w = h_2 - h_1$ = 450.4 − 408.2 = **42.3 kJ/kg**

$\dot{m}$ = Q / RE = 80 / 158.5 = **0.5047 kg/s**

$\dot{V} = \dot{m}\, v_1$ = 0.5047 × 0.0686 = **0.0346 m³/s**

COP = RE / w = 158.5 / 42.3 = **3.75**

Power = $\dot{m}$ w = 0.5047 × 42.3 = **21.33 kW**

Which table rows to read (R22, Tables A-7 / A-9)

1. Table A-7: row **-10 °C** → p_e; row **45 °C** → p_c; row **40 °C** → $h_f = h_3$.
2. Table A-9: p_e = 3.548 bar lies between the 3.5 and 4 bar blocks. In each block read the row **t_1 = 0 °C** (h, s, v), then interpolate to p_e.

3. Table A-9: p_c = 17.292 bar lies between the 16 and 18 bar blocks. In each block find the two rows whose s is either side of s_1 = 0.9690 (table basis), interpolate h at s_1, then interpolate between the bl… at p_c → h_2.

Table and chart use different enthalpy zeros (shift ≈ 154.9 kJ/kg). Read all h values from the same source. RE, w and COP do not change.

Property data: CoolProp 8.0 grid (IIR reference), interpolated in the page; works offline.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/R22-2024.html** in a browser.

D. UNDERSTAND Property-table rows used (how to read the table)

Table page not shown in the public copy. The scanned property-table page (Moran et al., Tables A-7 to A-15) is copyrighted. Open the same table in your own copy or the one printed with the exam; the rows to read and their values are listed below.

Table A-7 (saturated R22). Red: rows −10 °C → p_e = 3.5485 bar and 45 °C → p_c = 17.298 bar. Blue: row 40 °C → h_f = 94.53 kJ/kg = h_3.

Table page not shown in the public copy. The scanned property-table page (Moran et al., Tables A-7 to A-15) is copyrighted. Open the same table in your own copy or the one printed with the exam; the rows to read and their values are listed below.

Table A-9 (superheated R22), 3.5 bar and 4.0 bar blocks, row 0 °C (green) → v, h, s for state 1; interpolate to 3.5485 bar.

Table page not shown in the public copy. The scanned property-table page (Moran et al., Tables A-7 to A-15) is copyrighted. Open the same table in your own copy or the one printed with the exam; the rows to read and their values are listed below.

Table A-9, 16 bar block rows 70/80 °C and 18 bar block rows 80/90 °C (orange): the rows either side of s = 0.9685 → h_2 after interpolating to 17.298 bar.

Table basis (Moran A-7, A-9; h = 0 at −40 °C liquid)

p_e = 3.5485 bar, p_c = 17.298 bar [A-7]

State 1 (3.5485 bar, 0 °C): fraction = (3.5485 − 3.5)/(4.0 − 3.5) = 0.097
h_1 = 253.21 + 0.097(252.14 − 253.21) = **253.11**
s_1 = 0.9700 + 0.097(0.9542 − 0.9700) = **0.9685**
v_1 = 0.06956 + 0.097(0.06011 − 0.06956) = **0.0686** m³/kg [A-9]

State 2 (s = 0.9685):
16 bar: h = 287.30 + 0.687(295.93 − 287.30) = 293.23
18 bar: h = 293.40 + 0.368(302.16 − 293.40) = 296.62
at 17.298 bar: h_2 = 293.23 + 0.649(296.62 − 293.23) = **295.43** [A-9]

State 3: $h_3 = h_f$(40 °C) = **94.53** [A-7]

RE = $h_1 - h_4$ = 253.11 − 94.53 = **158.58 kJ/kg**
w = $h_2 - h_1$ = 295.43 − 253.11 = **42.32 kJ/kg**
COP = RE / w = 158.58 / 42.32 = **3.75**
m = Q / RE = 80 / 158.58 = **0.5045 kg/s**
V = m v_1 = 0.5045 × 0.0686 = **0.0346 m³/s**
P = m w = 0.5045 × 42.32 = **21.35 kW**

Chart basis (R22 P–h chart, IIR: h = 200 at 0 °C liquid)

p_e = 3.55 bar, p_c = 17.29 bar [chart]

State 1: read at 3.55 bar × 0 °C
h_1 = **408.2**, s_1 = 1.792, v_1 = **0.0686** m³/kg [chart]

State 2: follow s = 1.792 up to 17.29 bar
h_2 = **450.4** (t_2 ≈ 81 °C) [chart]

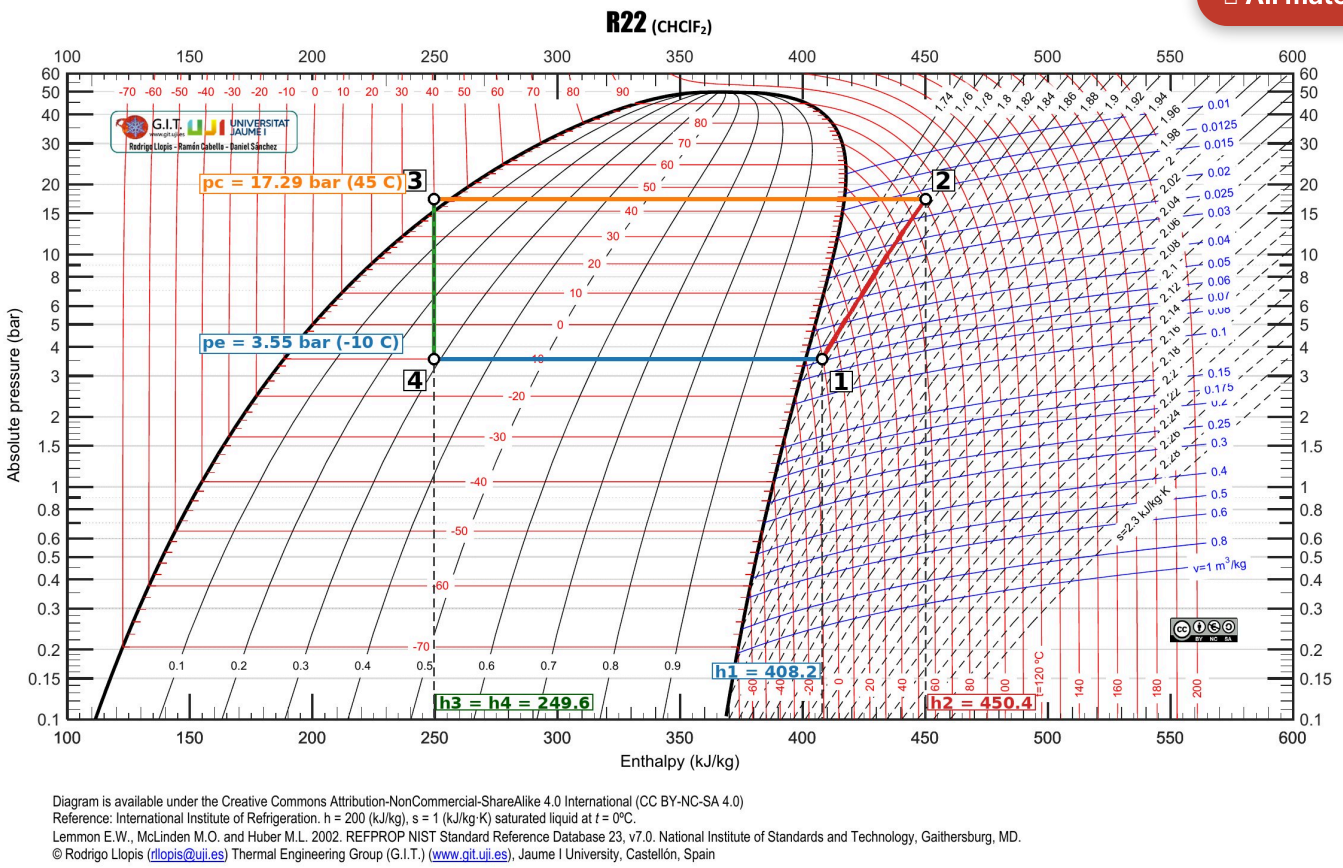
State 3: $h_3 = h_f$(40 °C) = **249.6** [chart]

RE = $h_1 - h_4$ = 408.2 − 249.6 = **158.6 kJ/kg**
w = $h_2 - h_1$ = 450.4 − 408.2 = **42.2 kJ/kg**
COP = RE / w = 158.6 / 42.2 = **3.76**
m = Q / RE = 80 / 158.6 = **0.5044 kg/s**
V = m v_1 = 0.5044 × 0.0686 = **0.0346 m³/s**
P = m w = 0.5044 × 42.2 = **21.29 kW**

All h values here are about 155.1 kJ/kg higher than the table column: only the zero is different.

Same answers on both sides: RE ≈ 158.6, w ≈ 42.2–42.3, COP ≈ 3.75, m ≈ 0.5045 kg/s, V ≈ 0.0346 m³/s, P ≈ 21.3 kW. Use **one** column only.

E. UNDERSTAND Whole P–h chart with the cycle



The whole R22 P–h chart with this cycle drawn on it. Dotted lines drop to the h-axis to read h_1, h_2, $h_3 = h_4$.

R22 p–h chart: © Rodrigo Llopis, Thermal Engineering Group (G.I.T.), Jaume I University, Castellón, Spain; CC BY-NC-SA 4.0; data from NIST REFPROP v7.0. Reference: IIR, h = 200 kJ/kg, s = 1 kJ/kg K for saturated liquid at 0 °C. Source: energiazero.org/cartelle/aermec/gas/R22_ph.pdf. Cycle overlay added for personal study.

F. WRITE IN EXAM Model answer

Given: R22, Q = 80 kW, t_e = −10 °C, t_c = 45 °C, superheat 10 K, subcool 5 K.

State points (from R22 P–h chart):

State	p (bar)	t (°C)	h (kJ/kg)	Source
1 (superheated vapour)	3.55	0	408.2; v_1 = 0.0686 m³/kg	chart
2 (after isentropic compression)	17.29	≈ 81	450.4	chart, $s_2 = s_1$
3 (subcooled liquid)	17.29	40	249.6	chart, h_f at 40 °C
4 (after throttling)	3.55	−10	249.6	$h_4 = h_3$

Refrigerating effect, RE = $h_1 - h_4$ = 408.2 − 249.6 = 158.6 kJ/kg

Compressor work, w = $h_2 - h_1$ = 450.4 − 408.2 = 42.2 kJ/kg

Mass flow, m = Q / RE = 80 / 158.6 = 0.5044 kg/s

(i) Volume flow rate (at compressor suction), V = m v_1 = 0.5044 × 0.0686 = **0.0346 m³/s**

(ii) COP = RE / w = 158.6 / 42.2 = **3.76**

(iii) Power, P = m w = 0.5044 × 42.2 = **21.3 kW**

P–h sketch checklist (draw it, 2 marks):

1. Saturation dome; two **horizontal** lines: p_c (top) and p_e (bottom).
2. Point 1 on the p_e line, **right of** the saturated-vapour curve (superheated).
3. 1 → 2 along a constant-s line up to p_c.
4. Point 3 on the p_c line, **left of** the saturated-liquid curve (subcooled).
5. 3 → 4 **vertical** ($h_4 = h_3$) down to p_e. Label h_1, h_2, h_3 on the h-axis.

If the exam gives Tables A-7/A-9 instead of the chart, write the left column of section D in the same table form: h_1 = 253.11, h_2 = 295.43, h_3 = 94.53, v_1 = 0.0686. Answers come out the same.

G. UNDERSTAND Common mistakes

- **Mixing tables with different h zeros.** h_1 from the chart (408) with h_3 from Table A-7 (94.5) gives RE = 314: wrong by double. Take every h from one source.
- **Volume flow at the wrong point.** "Volume flow rate of refrigerant" means at compressor suction, so use v_1 (0.0686), not v at 2 or v_g at −10 °C (0.0652).
- **Forgetting to interpolate.** 3.5485 bar and 17.298 bar are not printed blocks; read the blocks either side and interpolate. Skipping it costs about 1 kJ/kg on h_2.
- **Using the saturation row for state 1.** With 10 K superheat, state 1 is at 0 °C in the superheated table (h = 253.1), not h_g(−10 °C) = 246.15.
- **Using h_f(45 °C) for state 3.** 5 K subcooling means 40 °C liquid.
- **Rounding.** 158.6/42.2 = 3.76; with unrounded values COP = 3.75. Both get full marks.

2. R22 standard VCR cycle using the exam table: 2020 2(c)

Topic A4 · 12 marks ·

A. Problem

A standard VCR system developing 50 kW of refrigeration using R22 operates with a condensing temperature of 35 °C and an evaporating temperature of −10 °C. The enthalpy of the refrigerant after compression is 435.2 kJ/kg. Using the table, determine: (i) the refrigerating effect in kJ/kg, (ii) the circulation rate of refrigerant in kg/s, (iii) the power required by the compressor in kW, and (iv) COP.

B. UNDERSTAND How the cycle/process works

For understanding – do not write

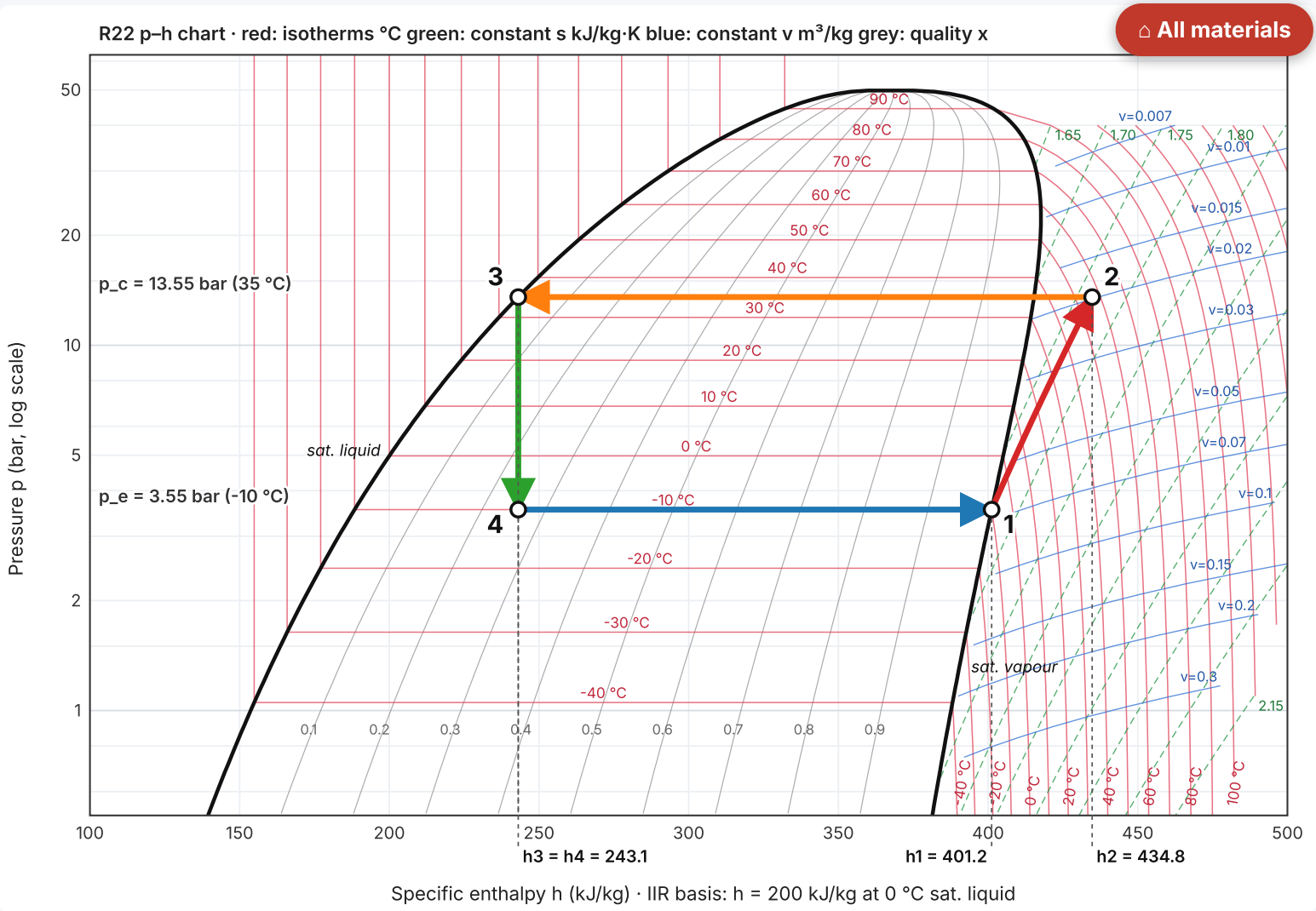
"Standard" cycle = no superheat and no subcooling. So point 1 is **saturated vapour** at −10 °C (on the right curve of the dome) and point 3 is **saturated liquid** at 35 °C (on the left curve). h_2 is given, so no entropy work is needed.

Point	State	Crossing of	h from
1	sat. vapour, −10 °C	p_e line × sat. vapour curve	h_g at −10 °C
2	superheated vapour	p_c line × constant-s line through 1	given 435.2
3	sat. liquid, 35 °C	p_c line × sat. liquid curve	h_f at 35 °C
4	wet mixture	vertical from 3 × p_e line	$h_4 = h_3$

The table has no −10 °C or 35 °C row. It gives −12/−8 and 34/36 °C. Each target is exactly in the middle, so the value is just the average of the two rows (linear interpolation).

Use **only** the table printed with the question. The Moran table (A-7) has a different enthalpy zero and does not match the given h_2 = 435.2.

C. UNDERSTAND Interactive graph



Hover over the chart to read h, p, t, s and v at any point. Press **Step through** to follow the refrigerant round the cycle.

p_e	**3.548**	bar	p_c	**13.548**	bar	h_1	**401.2**	kJ/kg
h_2	**434.8**	kJ/kg	$h_3 = h_4$	**243.1**	kJ/kg	v_1	**0.0653**	m³/kg
t_2 (discharge)	**56.8**	°C	x_4		**0.257**	RE = $h_1 - h_4$	**158.1**	kJ/kg
$w = h_2 - h_1$	**33.6**	kJ/kg	$\dot{m} = Q/RE$	**0.3162**	kg/s	$\dot{V} = \dot{m}\, v_1$	**0.0206**	m³/s
COP = RE/w	**4.71**		Power = $\dot{m}$ w	**10.62**	kW	Q_{cond}	**60.6**	kW
p_c/p_e	**3.82**							

Working (formula = substitution = answer)

RE = $h_1 - h_4$ = 401.2 − 243.1 = **158.1 kJ/kg**

w = $h_2 - h_1$ = 434.8 − 401.2 = **33.6 kJ/kg**

$\dot{m}$ = Q / RE = 50 / 158.1 = **0.3162 kg/s**

$\dot{V} = \dot{m}\, v_1$ = 0.3162 × 0.0653 = **0.0206 m³/s**

COP = RE / w = 158.1 / 33.6 = **4.71**

Power = $\dot{m}$ w = 0.3162 × 33.6 = **10.62 kW**

Which table rows to read (R22, Tables A-7 / A-9)

1. Table A-7: row **-10 °C** → p_e, $h_g = h_1$, s_g; row **35 °C** → p_c; row **35 °C** → $h_f = h_3$. *(35 °C is not printed: interpolate between 32 and 36 °C; 35 °C is not printed: interpolate between 32 and 36 °C)*

2. Table A-9: p_c = 13.548 bar lies between the 12 and 14 bar blocks. In each block find the two rows whose s is either side of s_1 = 0.9431 (table basis), interpolate h at s_1, then interpolate between the blo… p_c h_2.

Table and chart use different enthalpy zeros (shift ≈ 154.9 kJ/kg). Read all h values from the same source. RE, w and COP do not change.

Property data: CoolProp 8.0 grid (IIR reference), interpolated in the page; works offline.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/R22-2020.html** in a browser.

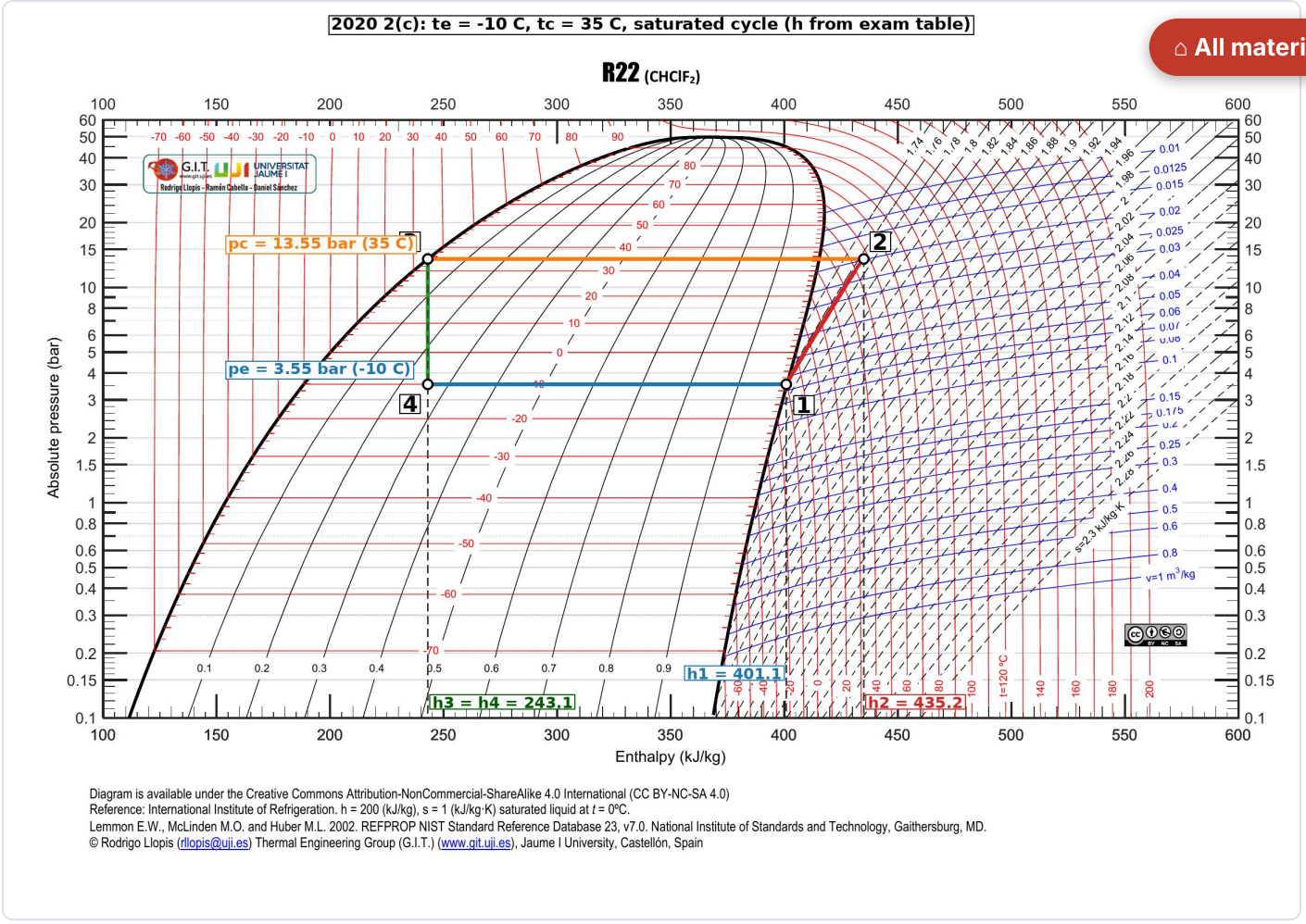
D. UNDERSTAND Property-table rows used (how to read the table)

Table: Properties of saturated liquid and saturated vapor of R22

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Entropy (kJ/kgK)	
		Liquid	Vapor	Liquid	Vapor
−12	3.3040	186.09	400.27	0.9484	1.7686
−8	3.8059	190.69	401.85	0.9657	1.7621
34	13.215	241.84	414.79	1.1419	1.7050
36	13.898	244.44	415.18	1.1501	1.7024

Table printed with the 2020 question. Blue: rows −12 and −8 °C, vapour h → average = h_1. Red: rows 34 and 36 °C, liquid h → average = h_3.

E. UNDERSTAND Whole P–h chart with the cycle



The 2020 cycle drawn on the full R22 P–h chart. The exam table uses the same IIR enthalpy zero as this chart, so the numbers line up.

R22 p–h chart: © Rodrigo Llopis, Thermal Engineering Group (G.I.T.), Jaume I University, Castellón, Spain; CC BY-NC-SA 4.0; data from NIST REFPROP v7.0. Reference: IIR, h = 200 kJ/kg, s = 1 kJ/kg K for saturated liquid at 0 °C. Source: energiazero.org/cartelle/aermec/gas/R22_ph.pdf. Cycle overlay added for personal study.

F. WRITE IN EXAM Model answer

Given: R22, Q = 50 kW, t_e = −10 °C, t_c = 35 °C, h_2 = 435.2 kJ/kg, standard cycle.

State	t (°C)	h (kJ/kg)	Source
1 (sat. vapour)	−10	(400.27 + 401.85)/2 = 401.06	given table, −12 and −8 °C rows
2	–	435.2	given
3 (sat. liquid)	35	(241.84 + 244.44)/2 = 243.14	given table, 34 and 36 °C rows
4	−10	243.14	$h_4 = h_3$ (throttling)

(i) RE = $h_1 - h_4$ = 401.06 − 243.14 = **157.92 kJ/kg**

(ii) m = Q / RE = 50 / 157.92 = **0.3166 kg/s**

(iii) w = $h_2 - h_1$ = 435.2 − 401.06 = 34.14 kJ/kg; P = m w = 0.3166 × 34.14 = **10.81 kW**

(iv) COP = RE / w = 157.92 / 34.14 = **4.63**

P–h sketch checklist (draw it, 2 marks):

1. Saturation dome; two **horizontal** lines: p_c (top) and p_e (bottom).
2. Point 1 on the p_e line, **on** the saturated-vapour curve (standard cycle).
3. 1 → 2 along a constant-s line up to p_c.
4. Point 3 on the p_c line, **on** the saturated-liquid curve.
5. 3 → 4 **vertical** ($h_4 = h_3$) down to p_e. Label h_1, h_2, h_3 on the h-axis.

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Using a different table.** The question says "using the table": Moran A-7 gives h_f(35 °C) ≈ 87, which does not fit h_2 = 435.2.
- **Swapping columns.** h_1 comes from the **vapour** column at the evaporator temperature; h_3 from the **liquid** column at the condenser temperature.
- **Not interpolating.** Taking the −12 °C row as −10 °C changes h_1 by 0.8 kJ/kg.
- **Power in kJ/s.** kg/s × kJ/kg = kJ/s = kW, so no extra conversion is needed.

3. Ammonia ice plant, dry saturated after compression: 2023 2(c)

Topic A11 · 12 marks ·

A. Problem

An ammonia refrigerator produces 25 tonnes of ice from and at 0 °C in 24 hours. The temperature range of the compressor is from 15 °C to −5 °C. The vapour is dry saturated at the end of compression and an expansion valve is used. Assume a COP of 45% of the theoretical value. Calculate the power required to drive the compressor. Latent heat of ice is 335 kJ/kg.

B. UNDERSTAND How the cycle/process works

For understanding – do not write

"Dry saturated at the end of compression" fixes point **2**, not point 1: point 2 sits **on** the saturated-vapour curve at the condenser temperature. Compression is isentropic, so point 1 has the same entropy, which puts it **inside** the dome (wet vapour) at the evaporator temperature. That is why you find a dryness fraction x_1 from entropy first.

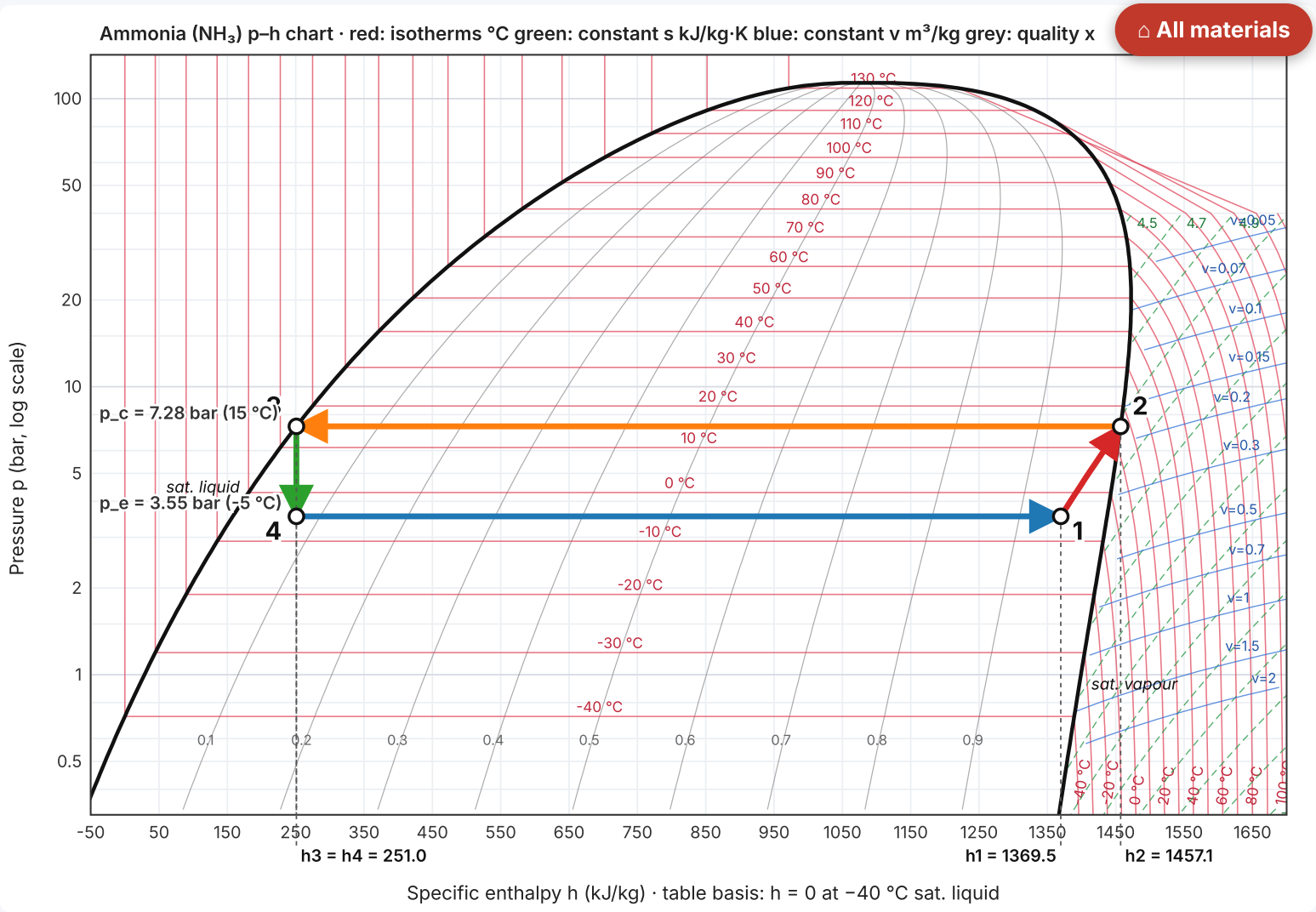
Point	State	Crossing of
2	dry saturated vapour at t_c	p_c line × saturated-vapour curve
1	wet vapour at t_e, $x_1 < 1$	p_e line × constant-s line through 2
3	saturated liquid at t_c	p_c line × saturated-liquid curve
4	wet mixture at t_e	vertical from 3 × p_e line

Ice load: making ice "from and at 0 °C" needs only the latent heat (335 kJ/kg); there is no sensible part. Divide by 86 400 s to get kW.

Actual COP = given percentage × theoretical COP. Power = load / actual COP. (The interactive graph shows the *theoretical* power m w; divide it by the percentage to get the real answer.)

Interpolation here: Table A-13 prints 12 and 16 °C (15 °C is 3/4 of the way: fraction 0.75) and −6 and −4 °C (−5 °C is the midpoint).

C. UNDERSTAND Interactive graph



Hover over the chart to read h, p, t, s and v at any point. Press **Step through** to follow the refrigerant round the cycle.

p_e	**3.547**	bar	p_c	**7.282**	bar	h_1	**1369.5**	kJ/kg
h_2	**1457.1**	kJ/kg	$h_3 = h_4$	**251.0**	kJ/kg	v_1	**0.3283**	m³/kg
t_2 (discharge)	**15.0**	°C	x_1 (wet suction)		**0.947**	RE = $h_1 - h_4$	**1118.5**	kJ/kg
$w = h_2 - h_1$	**87.6**	kJ/kg	$\dot{m} = Q/RE$	**0.0866**	kg/s	$\dot{V} = \dot{m}\, v_1$	**0.0284**	m³/s
COP = RE/w		**12.77**	Power = $\dot{m}\, w$	**7.59**	kW	Q_{cond}	**104.5**	kW
p_c/p_e		**2.05**						

Working (formula = substitution = answer)

RE = $h_1 - h_4$ = 1369.5 − 251.0 = **1118.5 kJ/kg**

$w = h_2 - h_1$ = 1457.1 − 1369.5 = **87.6 kJ/kg**

$\dot{m}$ = Q / RE = 96.9 / 1118.5 = **0.0866 kg/s**

$\dot{V} = \dot{m}\, v_1$ = 0.0866 × 0.3283 = **0.0284 m³/s**

COP = RE / w = 1118.5 / 87.6 = **12.77**

Power = $\dot{m}$ w = 0.0866 × 87.6 = **7.59 kW**

Which table rows to read (Ammonia, Tables A-13 / A-15)

1. Table A-13: row **-5 °C** → p_e, $h_g = h_1$, s_g; row **15 °C** → p_c; row **15 °C** → $h_f = h_3$. *(-5 °C is not printed: interpolate between -6 and -4 °C; 15 °C is not printed: interpolate between 12 and 16 °C; 15 °C is not printed: interpolate between 12 and 16 °C)*

2. Table A-15: p_c = 7.282 bar lies between the 7 and 8 bar blocks. In each block find the two rows whose s is either side of s_1 = 5.1476 (table basis), interpolate h at s_1, then interpolate between the blocks at p_c = 7.282 bar.

Table and chart use different enthalpy zeros (shift ≈ 18.9 kJ/kg). Read all h values from the same source. RE, w and COP do not change.

Property data: CoolProp 8.0 grid (IIR reference), interpolated in the page; works offline.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/NH3-2023.html** in a browser.

D. UNDERSTAND Property-table rows used (how to read the table)

Table page not shown in the public copy. The scanned property-table page (Moran et al., Tables A-7 to A-15) is copyrighted. Open the same table in your own copy or the one printed with the exam; the rows to read and their values are listed below.

Table A-13 (saturated ammonia). Blue: rows −6 and −4 °C → h_f, h_g, s_f, s_g at −5 °C (midpoint). Red: rows 12 and 16 °C → h_f, h_g, s_g at 15 °C (fraction 0.75).

F. WRITE IN EXAM Model answer

Given: NH_3, 25 t of ice per day from and at 0 °C, t_c = 15 °C, t_e = -5 °C, dry saturated after compression, COP = 45% of theoretical, L = 335 kJ/kg.

Property (Table A-13)	at 15 °C	at -5 °C
h_f (kJ/kg)	250.44	156.86
h_g (kJ/kg)	1456.04	1436.42
s_f, s_g (kJ/kg K)	s_g = 5.1433	0.6248, 5.3967
15 °C = 12 + 0.75(16 − 12): interpolated between the 12 and 16 °C rows. −5 °C = average of the −6 and −4 °C rows.		

State 2 is dry saturated at 15 °C: $h_2 = h_g = 1456.04$ kJ/kg, $s_2 = 5.1433$

1 → 2 isentropic: $s_1 = s_2 \Rightarrow x_1 = (s_2 - s_{f1})/(s_{g1} - s_{f1}) = (5.1433 - 0.6248)/(5.3967 - 0.6248) = 0.9469$

$h_1 = h_{f1} + x_1(h_{g1} - h_{f1}) = 156.86 + 0.9469(1436.42 - 156.86) = 1368.48$ kJ/kg

$h_4 = h_3 = h_f(15\ °C) = 250.44$ kJ/kg

$COP_{th} = (h_1 - h_4)/(h_2 - h_1) = (1368.48 - 250.44)/(1456.04 - 1368.48) = 12.77$

$COP_{actual} = 0.45 \times 12.77 = 5.746$

Refrigeration load Q = (ice per day × L) / seconds per day = 25 × 1000 × 335 / (24 × 3600) = 96.93 kW

Power = Q / COP_{actual} = 96.93 / 5.746 = **16.87 kW**

Sketch (T–s or P–h): point 1 **inside** the dome (wet, $x_1 \approx 0.95$), 1 → 2 vertical on T–s ending **on** the saturated-vapour curve at 15 °C, 3 on the saturated-liquid curve, 3 → 4 constant h.

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Putting point 1 on the vapour curve.** Here point **2** is dry saturated; point 1 is wet ($x_1 \approx 0.95$).
- **Adding sensible heat to the ice load.** "From and at 0 °C" means latent heat only.
- **Applying 45% the wrong way.** Actual COP is *smaller* than theoretical, so power is *bigger*: P = Q/(0.45 COP_{th}).
- **Table zero.** Other ammonia tables (e.g. the course slides) use a different zero and give h values about 1 kJ/kg different; COP and power come out the same to 3 figures (the old pack got 16.9 kW).

4. Ammonia ice plant, dry saturated after compression: 2021 2(c)

Topic A11 · practice ·

A. Problem

An ammonia ice plant produces 30 tonnes of ice per day from and at 0 °C. The condenser and evaporator temperatures are 20 °C and −10 °C. The vapour is dry saturated at the end of compression. The actual COP is 55% of the theoretical value. Find the power required to drive the compressor. (Latent heat of ice 335 kJ/kg.)

B. UNDERSTAND How the cycle/process works

For understanding – do not write

"Dry saturated at the end of compression" fixes point **2**, not point 1: point 2 sits **on** the saturated-vapour curve at the condenser temperature. Compression is isentropic, so point 1 has the same entropy, which puts it **inside** the dome (wet vapour) at the evaporator temperature. That is why you find a dryness fraction x_1 from entropy first.

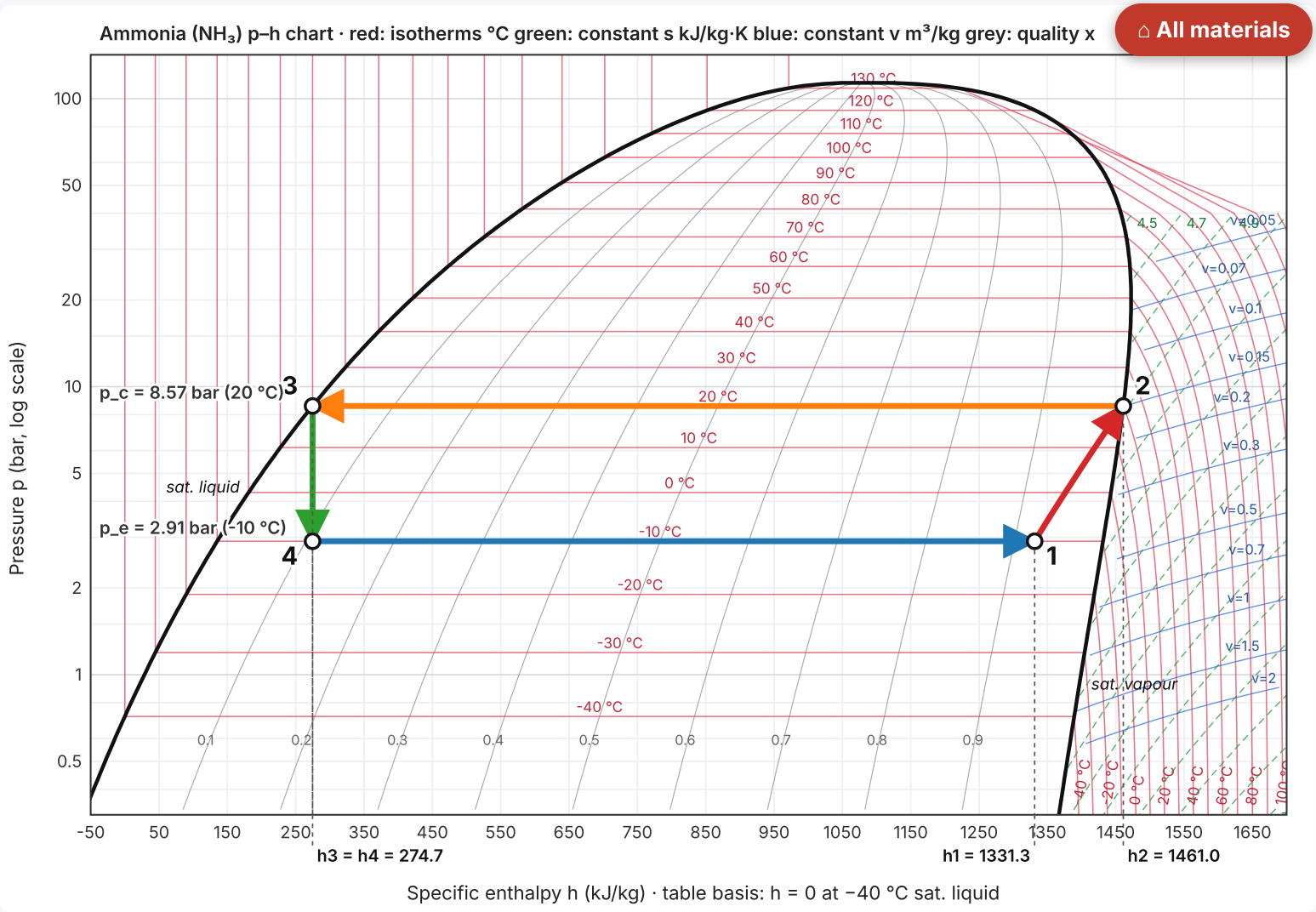
Point	State	Crossing of
2	dry saturated vapour at t_c	p_c line × saturated-vapour curve
1	wet vapour at t_e, $x_1 < 1$	p_e line × constant-s line through 2
3	saturated liquid at t_c	p_c line × saturated-liquid curve
4	wet mixture at t_e	vertical from 3 × p_e line

Ice load: making ice "from and at 0 °C" needs only the latent heat (335 kJ/kg); there is no sensible part. Divide by 86 400 s to get kW.

Actual COP = given percentage × theoretical COP. Power = load / actual COP. (The interactive graph shows the *theoretical* power m w; divide it by the percentage to get the real answer.)

Both 20 °C and −10 °C are printed rows in Table A-13, so no interpolation is needed.

C. UNDERSTAND Interactive graph



Hover over the chart to read h, p, t, s and v at any point. Press **Step through** to follow the refrigerant round the cycle.

p_e	**2.906**	bar	p_c	**8.570**	bar	h_1	**1331.3**	kJ/kg
h_2	**1461.0**	kJ/kg	$h_3 = h_4$	**274.7**	kJ/kg	v_1	**0.3860**	m³/kg
t_2 (discharge)	**20.0**	°C	x_1 (wet suction)		**0.923**	RE = $h_1 - h_4$	**1056.6**	kJ/kg
$w = h_2 - h_1$	**129.7**	kJ/kg	$\dot{m} = Q/RE$	**0.1101**	kg/s	$\dot{V} = \dot{m}\, v_1$	**0.0425**	m³/s
COP = RE/w	**8.15**		Power = $\dot{m}\, w$	**14.28**	kW	Q_{cond}	**130.6**	kW
p_c/p_e	**2.95**							

Working (formula = substitution = answer)

RE = $h_1 - h_4$ = 1331.3 − 274.7 = **1056.6 kJ/kg**

$w = h_2 - h_1$ = 1461.0 − 1331.3 = **129.7 kJ/kg**

$\dot{m}$ = Q / RE = 116.3 / 1056.6 = **0.1101 kg/s**

$\dot{V} = \dot{m}\, v_1$ = 0.1101 × 0.3860 = **0.0425 m³/s**

COP = RE / w = 1056.6 / 129.7 = **8.15**

Power = $\dot{m}$ w = 0.1101 × 129.7 = **14.28 kW**

Which table rows to read (Ammonia, Tables A-13 / A-15)

1. Table A-13: row **-10 °C** → p_e, $h_g = h_1$, s_g; row **20 °C** → p_c; row **20 °C** → $h_f = h_3$.
2. Table A-15: p_c = 8.570 bar lies between the 8 and 9 bar blocks. In each block find the two rows whose s is either side of s_1 = 5.0894 (table basis), interpolate h at s_1, then interpolate between the blocks to p_c → h_2.

Table and chart use different enthalpy zeros (shift ≈ 18.9 kJ/kg). Read all h values from the same source. RE, w and COP do not change.

⌂ All materials

Property data: CoolProp 8.0 grid (IIR reference), interpolated in the page; works offline.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/NH3-2021.html** in a browser.

D. UNDERSTAND Property-table rows used (how to read the table)

Table page not shown in the public copy. The scanned property-table page (Moran et al., Tables A-7 to A-15) is copyrighted. Open the same table in your own copy or the one printed with the exam; the rows to read and their values are listed below.

Table A-13. Blue: row −10 °C → h_f, h_g, s_f, s_g. Red: row 20 °C → h_f, h_g, s_g.

F. WRITE IN EXAM Model answer

Given: NH_3, 30 t of ice per day from and at 0 °C, t_c = 20 °C, t_e = -10 °C, dry saturated after compression, COP = 55% of theoretical, L = 335 kJ/kg.

Property (Table A-13)	at 20 °C	at -10 °C
h_f (kJ/kg)	274.26	133.94
h_g (kJ/kg)	1459.90	1430.55
s_f, s_g (kJ/kg K)	s_g = 5.0849	0.5389, 5.4662
Both rows are printed in Table A-13.		

State 2 is dry saturated at 20 °C: $h_2 = h_g = 1459.90$ kJ/kg, $s_2 = 5.0849$

$1 \rightarrow 2$ isentropic: $s_1 = s_2 \Rightarrow x_1 = (s_2 - s_{f1})/(s_{g1} - s_{f1}) = (5.0849 - 0.5389)/(5.4662 - 0.5389) = 0.9226$

$h_1 = h_{f1} + x_1(h_{g1} - h_{f1}) = 133.94 + 0.9226(1430.55 - 133.94) = 1330.21$ kJ/kg

$h_4 = h_3 = h_f(20\ °C) = 274.26$ kJ/kg

$COP_{th} = (h_1 - h_4)/(h_2 - h_1) = (1330.21 - 274.26)/(1459.90 - 1330.21) = 8.14$

$COP_{actual} = 0.55 \times 8.14 = 4.478$

Refrigeration load Q = (ice per day × L) / seconds per day = 30 × 1000 × 335 / (24 × 3600) = 116.32 kW

Power = Q / COP_{actual} = 116.32 / 4.478 = **25.97 kW**

Sketch (T–s or P–h): point 1 **inside** the dome (wet, $x_1 \approx 0.92$), $1 \rightarrow 2$ vertical on T–s ending **on** the saturated-vapour curve at 20 °C, 3 on the saturated-liquid curve, $3 \rightarrow 4$ constant h.

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Putting point 1 on the vapour curve.** Here point **2** is dry saturated; point 1 is wet ($x_1 \approx 0.95$).
- **Adding sensible heat to the ice load.** "From and at 0 °C" means latent heat only.
- **Applying 45% the wrong way.** Actual COP is *smaller* than theoretical, so power is *bigger*: P = Q/(0.45 COP_{th}).

5. Refrigeration tonnage: ice and cold-storage loads (2024 1a, 2023 1d, 2020 1c)

Topic A3 · 10 + 7 + 10 marks ·

A. Problem

2024 1(a): Calculate the refrigeration tonnage for 2000 kg ice melting from 0 °C to 45 °C. **2023 1(d):** Calculate the refrigeration tonnage for 1500 kg ice from 0 °C to 25 °C. **2020 1(c):** 600 kg of fruit enters a cold store at 25 °C and is cooled to −5 °C in 8 hours; latent heat of freezing 105 kJ/kg, specific heat 1.255 kJ/kg K. Find the plant capacity.

B. UNDERSTAND How the cycle/process works

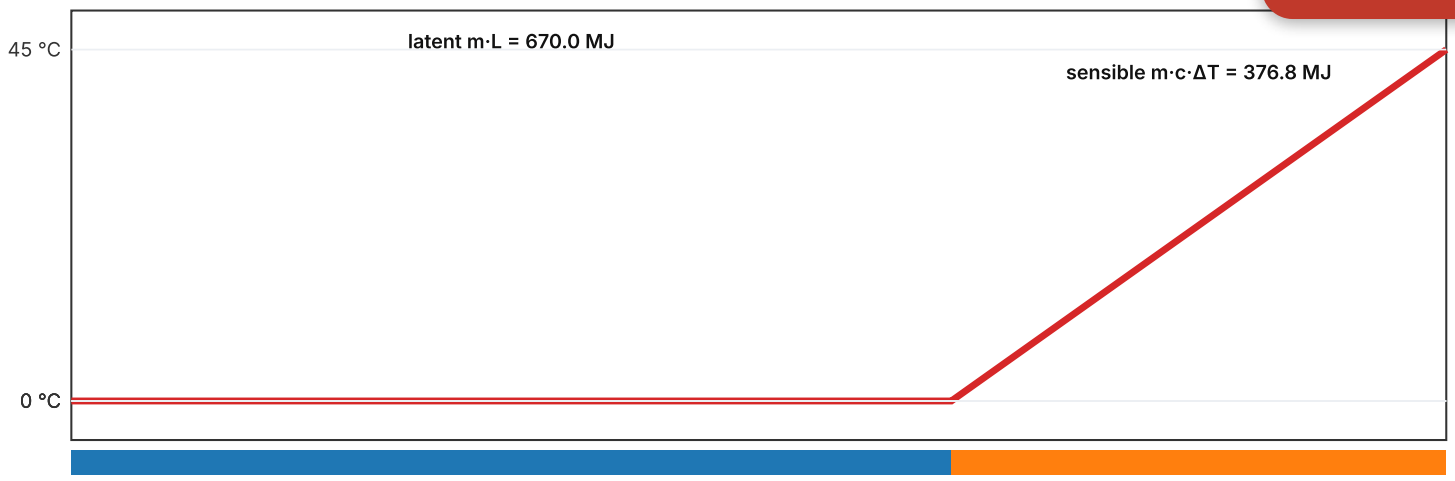
For understanding – do not write

1 TR (ton of refrigeration) = heat to melt 1 tonne of ice at 0 °C in 24 h ≈ **210 kJ/min = 3.5 kW**.

Any load = **latent part** (phase change, temperature constant: m L) + **sensible part** (temperature change: m c ΔT). Add them, divide by the time in minutes, then divide by 210.

Ice questions with no time given are taken over **24 h** (the definition of TR). Water c = 4.187 kJ/kg K, ice latent heat L = 335 kJ/kg.

C. UNDERSTAND Interactive graph



The flat part is the phase change (temperature stays the same); the sloped part is the temperature change. Total heat ÷ minutes ÷ 210 = TR.

Sensible	**376830**	kJ	Latent	**670000**	kJ	Total	**1046830**	kJ
Per minute	**727.0**	kJ/min	Capacity	**3.46**	TR	=	**12.12**	kW

Working (formula = substitution = answer)

Sensible = m c ΔT = 2000 × 4.187 × 45 = **376830 kJ**

Latent = m L = 2000 × 335 = **670000 kJ**

Total = 376830 + 670000 = **1046830 kJ**

Per minute = total / (hours × 60) = 1046830 / (24 × 60) = **727.0 kJ/min**

TR = (kJ/min) / 210 = 727.0 / 210 = **3.46 TR**

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/TR.html** in a browser.

F. WRITE IN EXAM Model answer

Write this in the exam

2024 1(a): Q = m[L + $c_w(t_2 - 0)$] = 2000[335 + 4.187 × 45] = 1,046,830 kJ per day

= 1,046,830 / 1440 = 727.0 kJ/min; TR = 727.0 / 210 = **3.46 TR**

2023 1(d): Q = 1500[335 + 4.187 × 25] = 659,512 kJ/day = 458.0 kJ/min; TR = **2.18 TR**

2020 1(c): Sensible = m c ΔT = 600 × 1.255 × (25 − (−5)) = 22,590 kJ; Latent = 600 × 105 = 63,000 kJ

Total = 85,590 kJ in 8 h = 85,590/480 = 178.3 kJ/min; Capacity = 178.3/210 = **0.85 TR**

Definition line (write it first, 1 mark): 1 TR = heat removed to freeze 1 tonne of ice from and at 0 °C in 24 h = 210 kJ/min ≈ 3.5 kW.

G. UNDERSTAND Common mistakes

⌂ All materials

For understanding – do not write

- **Forgetting the time.** TR is a rate: divide by minutes (24 h = 1440 min, 8 h = 480 min).
- **Using ice c for water.** Above 0 °C it is water: c = 4.187 kJ/kg K.
- **For the 2020 fruit:** the paper gives one c for the whole range, so use 1.255 × 30 K. Say so in one line.

6. Aircraft simple air-cycle refrigeration: 2020 3(c)

Topic A12 · 13 marks ·

A. Problem

In an aircraft refrigeration unit using the air cycle, 50 kg/min of air at 200 cm Hg gauge and 205 °C are bled off the air compressor serving the jet engine. The bled air passes through a heat exchanger leaving at 180 cm Hg gauge and 80 °C. It is then expanded through a small cooling turbine to 20 cm Hg vacuum and −10 °C. The air exhausted out of the plane is at 25 °C. Assume c_p = 1.0 kJ/kg K. Determine the cooling capacity in TR and the COP based on input power to the bled-off air.

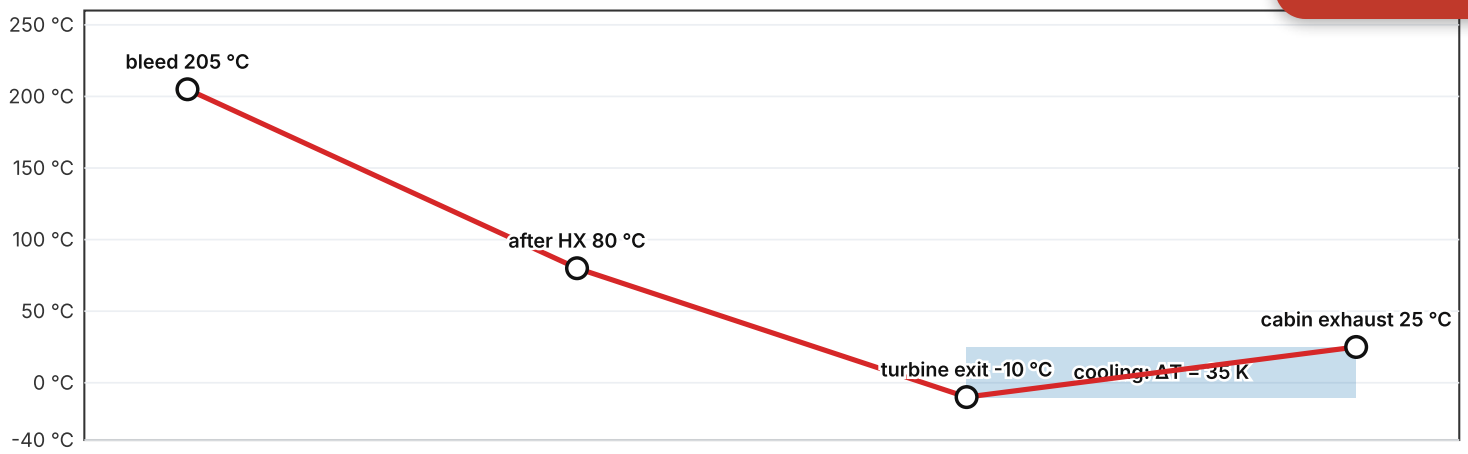
B. UNDERSTAND How the cycle/process works

For understanding – do not write

Air is the refrigerant. Hot bled air (205 °C) is cooled in the heat exchanger (80 °C), expanded in the cooling turbine to −10 °C, then sent to the cabin, where it warms up to 25 °C before leaving. **That warming in the cabin is the cooling it gives.**

The pressures (cm Hg) are not needed, because the temperatures at each point are given. The “input” is the energy carried by the bled air above the exhaust state: m c_p(205 − 25).

C. UNDERSTAND Interactive graph



Hot bled air is cooled in the heat exchanger, expanded in the turbine to below cabin temperature, then warms up in the cabin: that warming is the cooling it gives.

Refrigerating effect	**1750**	kJ/min	Capacity	**8.33**	TR	Input to bled air	**9000**	kJ/min
COP	**0.194**							

Working (formula = substitution = answer)

RE = $\dot{m}\, c_p(T_{cabin} - T_{turbine})$ = 50 × 1 × (25 − -10) = **1750 kJ/min**

Input = $\dot{m}\, c_p(T_{bleed} - T_{cabin})$ = 50 × 1 × (205 − 25) = **9000 kJ/min**

TR = RE / 210 = **8.33**; COP = RE / input = 1750 / 9000 = **0.194**

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/AIR.html** in a browser.

F. WRITE IN EXAM Model answer

Write this in the exam

Given: m = 50 kg/min, c_p = 1.0 kJ/kg K, bleed 205 °C, turbine exit −10 °C, cabin exhaust 25 °C.

Refrigerating effect = m $c_p(t_{exhaust} - t_{turbine})$ = 50 × 1.0 × (25 − (−10)) = 1,750 kJ/min

Cooling capacity = 1,750 / 210 = **8.33 TR**

Input to bled air = m $c_p(t_{bleed} - t_{exhaust})$ = 50 × 1.0 × (205 − 25) = 9,000 kJ/min

COP = refrigerating effect / input = 1,750 / 9,000 = **0.19**

Assumption stated: input energy measured from the cabin-exhaust state (25 °C).

G. UNDERSTAND Common mistakes

For understanding – do not write

⌂ All materials

- **Using the HX exit (80 °C) in the RE.** The cooling happens in the cabin, from −10 to 25 °C.
- **Converting the pressures.** Not needed here: all temperatures are given.
- **Low COP is normal** for air cycles (about 0.2). Do not “correct” it.

7. Summer air conditioning with ventilation air and bypass factor: 2024 5(c)

Topic B3 · 15 marks ·

A. Problem

A refrigeration (air-conditioning) plant is shown in the figure: room sensible heat 20 kW, room latent heat 5 kW, room 25 °C DBT / 50% RH, outside air 43 °C DBT / 27.5 °C WBT, return : outside air = 3 : 1 by mass, coil ADP 12 °C, bypass factor 0.13. Determine: (i) dehumidified air quantity, (ii) sensible and latent heat gain due to outside air, (iii) total refrigeration load of the plant.

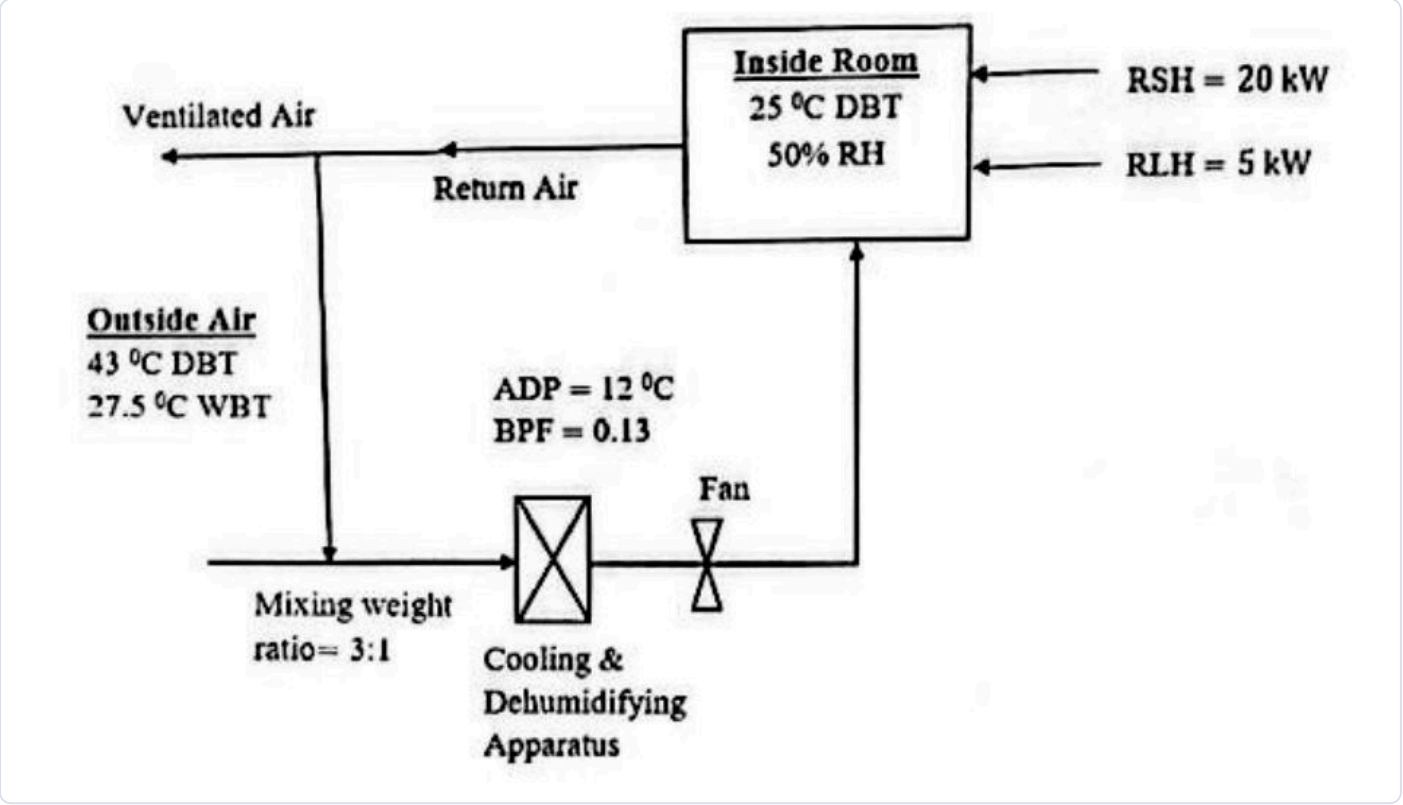


Figure given with the question

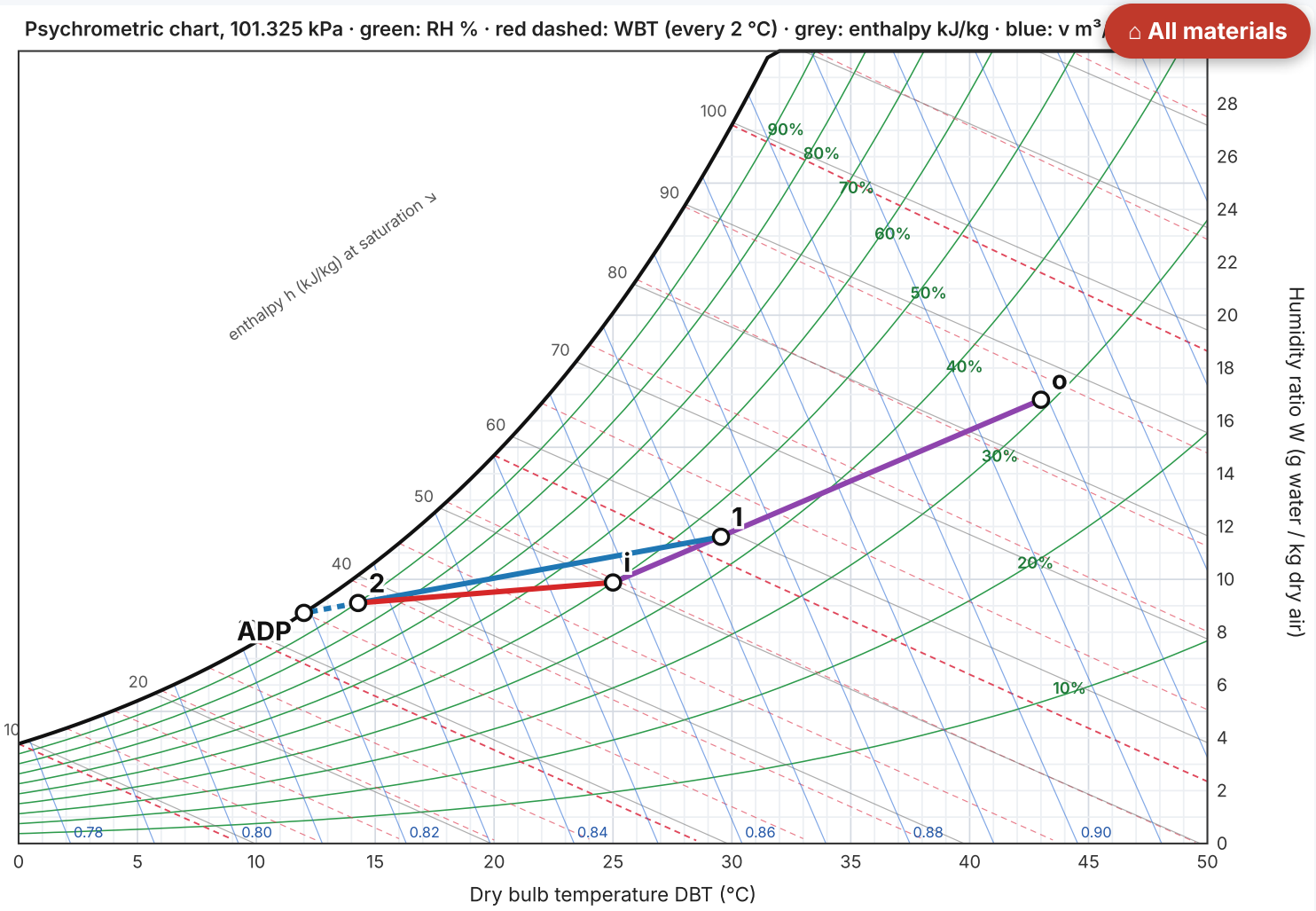
B. UNDERSTAND How the cycle/process works

Path of the air (follow it on the chart with Step through): outside air **o** mixes with room return air **i** → mixture **1** on the straight line o–i, 1/4 of the way from i (3:1 by mass) → cooling coil pulls it toward the **ADP** (apparatus dew point, coil surface temperature) but 13% of the air bypasses the coil, so it leaves at **2** → supply air 2 absorbs the room heat and becomes **i**.

Point	Found as the crossing of
i	DBT 25 °C vertical × RH 50% curve
o	DBT 43 °C vertical × WBT 27.5 °C sloping line
1	line o–i × DBT 29.5 °C (mass-weighted)
ADP	saturation curve × 12 °C
2	line 1–ADP × DBT $t_2 = ADP + X(t_1 - ADP)$

Dehumidified air = the air that goes through the coil = supply air. It must carry away the room sensible heat: $m_a = RSH / [c_p(t_i - t_2)]$. c_{pm} = 1.0216 kJ/kg K is humid specific heat (1.005 + 1.88 W); many books use 1.02.

C. UNDERSTAND Interactive graph



Hover to read any point (DBT, RH, WBT, DPT, W, h, v). Drag the sliders to see how the process moves.

State	DBT °C	WBT °C	DPT °C	RH %	W g/kg	h kJ/kg	v m³/kg
o (outside)	43.0	27.5	22.1	31	16.80	86.6	0.920
i (room)	25.0	17.9	13.9	50	9.88	50.3	0.858
1 (mix)	29.5	20.7	16.3	45	11.61	59.4	0.873
ADP	12.0	12.0	12.0	100	8.73	34.1	0.819
2 (supply)	14.3	13.3	12.6	90	9.10	37.4	0.826

ADP	**12.00**	°C	Dehumidified air $\dot{m}_a$	**1.826**	kg/s	Supply air	**90.5**	m³/min
Outside air $\dot{m}_o$	**0.457**	kg/s	Ventilation air	**25.2**	m³/min (cmm)	OASH	**8.40**	kW
OALH	**7.90**	kW	OATH	**16.57**	kW	GTH = RSH + RLH + OATH	**41.57**	kW
in TR	**11.82**	TR	Check: $\dot{m}_a(h_1 - h_2)$	**40.21**	kW	RSHF	**0.800**	

Working (formula = substitution = answer)

t_2 = ADP + X(t_1 − ADP) = 12.00 + 0.13(29.54 − 12.00) = **14.28 °C**

$\dot{m}_a$ = RSH / [1.0216(t_i − t_2)] = 20 / [1.0216(25 − 14.28)] = **1.826 kg/s**

$\dot{m}_o$ = fraction × $\dot{m}_a$ = 0.25 × 1.826 = **0.457 kg/s**

OASH = $\dot{m}_o$ × 1.0216 × (t_o − t_i) = 0.457 × 1.0216 × 18.0 = **8.40 kW**

OALH = $\dot{m}_o$ × 2501 × (W_o − W_i) = 0.457 × 2501 × 0.00692 = **7.90 kW**

OATH = $\dot{m}_o$(h_o − h_i) = 0.457(86.6 − 50.3) = **16.57 kW**

GTH = RSH + RLH + OATH = 20 + 5 + 16.57 = **41.57 kW**

F. WRITE IN EXAM Model answer

Write this in the exam

Given: RSH = 20 kW, RLH = 5 kW, room 25 °C / 50%, outside 43 °C DBT / 27.5 °C WBT, 3:1, ADP = 12 °C, X = 0.13.

State	DBT °C	W (g/kg)	h (kJ/kg)	Source
o	43	16.8	86.6	psy chart; v_o = 0.92 m³/kg
i	25	9.9	50.3	psy chart
1 (mix)	0.75(25) + 0.25(43) = 29.5	11.62	59.37	mass-weighted

Mixing by mass: $t_1 = 0.75\ t_i + 0.25\ t_o$, $W_1 = 0.75\ W_i + 0.25\ W_o$, $h_1 = 0.75\ h_i + 0.25\ h_o$

Coil exit: $t_2 = t_{ADP} + X(t_1 - t_{ADP}) = 12 + 0.13(29.5 - 12) = 14.28$ °C

(i) Dehumidified air, $m_a = RSH / [c_{pm}(t_i - t_2)] = 20 / [1.0216(25 - 14.28)] =$ **1.83 kg/s**

Outside air, $m_o = 0.25 \times 1.83 = 0.456$ kg/s

(ii) $OASH = m_o\ c_{pm}(t_o - t_i) = 0.456 \times 1.0216 \times 18 =$ **8.39 kW**

$OALH = m_o\ h_{fg}(W_o - W_i) = 0.456 \times 2501 \times (16.8 - 9.9)/1000 =$ **7.88 kW**

(iii) $OATH = m_o(h_o - h_i) = 0.456(86.6 - 50.3) = 16.57$ kW

Total load = RSH + RLH + OATH = 20 + 5 + 16.57 = **41.6 kW = 11.8 TR**

Chart sketch: mark o, i, join; 1 on it; ADP on saturation curve; line 1–ADP with 2 on it; line 2–i.

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Mixing by temperature only.** Mix W and h by mass; t_1 by mass is fine as an approximation.
- **Taking supply air = ADP.** With X = 0.13 the air leaves above ADP: t_2 = ADP + X(t_1 − ADP).
- **Forgetting the outside-air load.** The plant must handle room load + outside air load (GTH = RTH + OATH).
- **Chart reading.** Readings differ by 2–3% between charts (the older pack used W_o = 17 g/kg and got 41.8 kW). The method gets the marks.

8. Finding ADP from the room heat ratio: 2021 5(c)

Topic B3 · 15 marks ·

A. Problem

The room sensible heat gain and latent heat gain of a conditioned space are 40 kW and 15 kW. The inside design condition is 25 °C DBT and 55% RH. The return air is mixed with outside air before entering the cooling coil in the ratio 3:1 by weight. Assuming the bypass factor of the cooling coil is 0.13, determine: (i) apparatus dew point (ADP), (ii) ventilation air in cmm, (iii) total refrigeration load of the plant. *(Outside condition is not printed on some copies; taken as 35 °C DBT, 65% RH. State this.)*

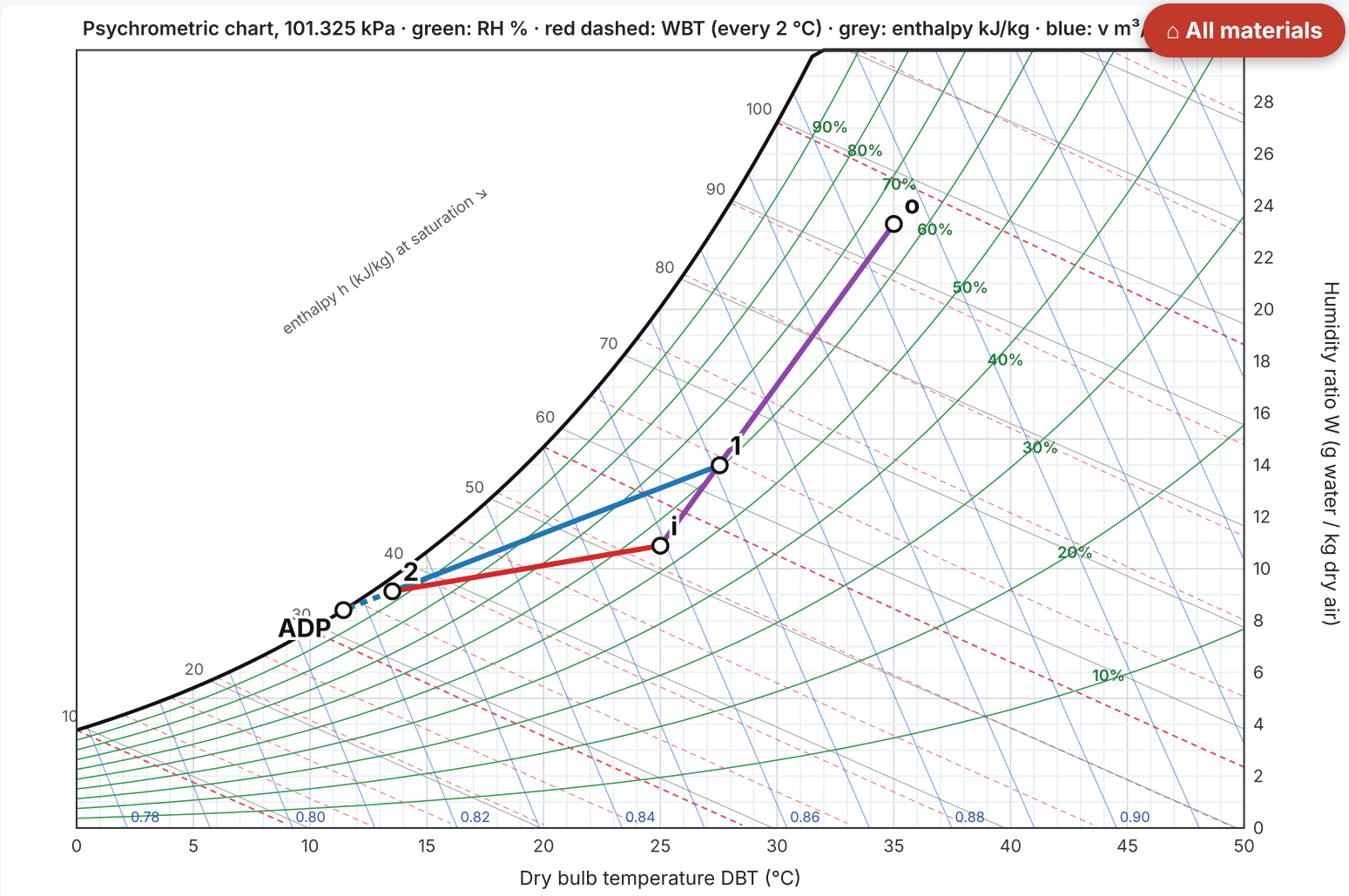
B. UNDERSTAND How the cycle/process works

For understanding – do not write

Here the ADP is **not given**. It is fixed by the room: the supply air 2 must lie on the room line (RSHF line) through i, whose slope is set by RSH/RLH = 40/15. It must also lie on the coil line from 1 to the ADP, at the bypass position. Only one ADP satisfies both, so you **try values** (or use the RSHF line on the chart: draw it from i, extend it to the saturation curve... but with bypass the clean way is trial).

Tick **"find ADP from RSH/RLH"** in the chart below to see it solved; untick it and move the ADP slider to watch the ratio change.

C. UNDERSTAND Interactive graph



Hover to read any point (DBT, RH, WBT, DPT, W, h, v). Drag the sliders to see how the process moves.

State	DBT °C	WBT °C	DPT °C	RH %	W g/kg	h kJ/kg	v m³/kg
o (outside)	35.0	29.1	27.4	65	23.29	95.0	0.906
i (room)	25.0	18.7	15.3	55	10.89	52.9	0.859
1 (mix)	27.5	21.8	19.2	61	13.99	63.4	0.871
ADP	11.4	11.4	11.4	100	8.40	32.7	0.817
2 (supply)	13.5	13.0	12.7	95	9.13	36.7	0.824

ADP	**11.43**	°C	Dehumidified air $\dot{m}_a$	**3.412**	kg/s	Supply air	**168.7**	m³/min
Outside air $\dot{m}_o$	**0.853**	kg/s	Ventilation air	**46.3**	m³/min (cmm)	OASH	**8.71**	kW
OALH	**26.47**	kW	OATH	**35.91**	kW	GTH = RSH + RLH + OATH	**90.91**	kW
in TR	**25.85**	TR	Check: $\dot{m}_a(h_1 - h_2)$	**91.24**	kW	RSHF		**0.727**

Working (formula = substitution = answer)

t_2 = ADP + X(t_1 − ADP) = 11.43 + 0.13(27.54 − 11.43) = **13.52 °C**

$\dot{m}_a$ = RSH / [1.0216(t_i − t_2)] = 40 / [1.0216(25 − 13.52)] = **3.412 kg/s**

$\dot{m}_o$ = fraction × $\dot{m}_a$ = 0.25 × 3.412 = **0.853 kg/s**

OASH = $\dot{m}_o$ × 1.0216 × (t_o − t_i) = 0.853 × 1.0216 × 10.0 = **8.71 kW**

OALH = $\dot{m}_o$ × 2501 × (W_o − W_i) = 0.853 × 2501 × 0.01241 = **26.47 kW**

OATH = $\dot{m}_o$(h_o − h_i) = 0.853(95.0 − 52.9) = **35.91 kW**

GTH = RSH + RLH + OATH = 40 + 15 + 35.91 = **90.91 kW**

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/AC-2021.html** in a browser.

F. WRITE IN EXAM Model answer

Write this in the exam

Given: RSH 40 kW, RLH 15 kW, room 25 °C / 55%, outside 35 °C / 65% (assumed), 3:1, X = 0.13.

State	DBT °C	W (g/kg)	h (kJ/kg)	Source
o	35	23.3	95.0	psy chart; v_o = 0.906
i	25	10.9	52.9	psy chart
1 (mix)	27.5	14.00	–	mass-weighted

(i) Conditions for 2: t_2 = ADP + 0.13(t_1 − ADP), $W_2 = W_{ADP} + 0.13(W_1 - W_{ADP})$, and $1.0216(t_i - t_2) / [2501(W_i - W_2)]$ = RSH/RLH = 2.67

Try ADP = 11.43 °C (W_{ADP} = 8.4 g/kg, psy chart): t_2 = 13.52 °C, W_2 = 9.13 g/kg → ratio = 1.0216(11.48) / [2501 × 0.0018] = 2.65 ≈ 2.67 ✓

ADP = **11.43 °C ≈ 11.4 °C**

m_a = 40 / [1.0216(25 − 13.52)] = 3.410 kg/s; m_o = 0.25 m_a = 0.853 kg/s

(ii) Ventilation air = $m_o v_o \times 60$ = 0.853 × 0.906 × 60 = **46.3 m³/min (cmm)**

(iii) OATH = $m_o(h_o - h_i)$ = 0.853(95.0 − 52.9) = 35.9 kW; Total = 40 + 15 + 35.9 = **90.9 kW = 25.8 TR**

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Not stating the assumed outside condition.** Write it in the first line.
- **Stopping after one trial.** Show one that works (ratio ≈ 2.67) and box the ADP.
- **cmm** = cubic metres per minute: multiply m^3/s by 60, and use v of **outside** air.

9. Ventilation rate from CO_2 balance: 2024 6(c)

Topic B5 · 15 marks ·

A. Problem

Dhaka has twice the CO_2 concentration of Khulna (400 ppm). If a classroom with 30 students is in a room of the same size in both cities, determine the required ventilation rate for each location. Also explain how to control the indoor air quality.

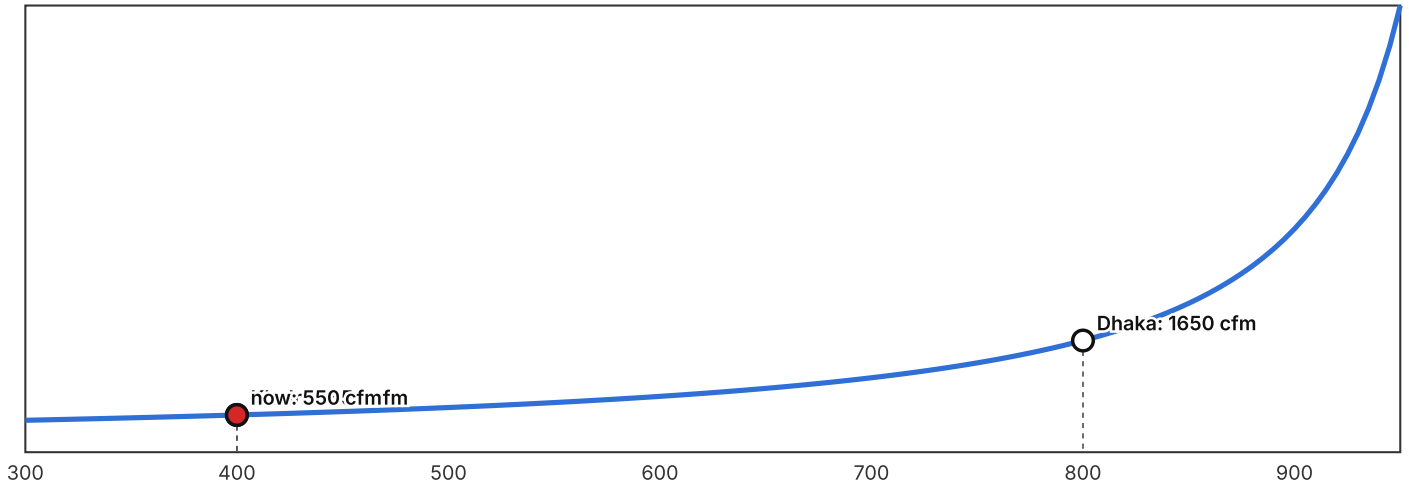
B. UNDERSTAND How the cycle/process works

For understanding – do not write

CO_2 balance in steady state: CO_2 brought in with fresh air + CO_2 breathed out = CO_2 leaving with exhaust air. So $V\,C_{out} + N\,N_p = V\,C_{in}$, giving $\mathbf{V = N\,N_p / (C_{in} - C_{out})}$.

The paper gives only the outdoor ppm, so take from the class notes: N_p = 0.011 cfm CO_2 per seated student, allowable indoor C_{in} = 1000 ppm. The dirtier the outdoor air, the smaller the margin (1000 − C_{out}), so the more fresh air you need.

C. UNDERSTAND Interactive graph



Mass balance: CO_2 in with fresh air + CO_2 breathed out = CO_2 leaving. So $V = N \cdot N_p / (C_{in} - C_{out})$.

CO_2 made	**0.330**	cfm	Fresh air	**550**	cfm	per person	**18.3**	cfm
in SI	**260**	L/s					

Working (formula = substitution = answer)

$V = N\,N_p / (C_{in} - C_{out}) = 30 \times 0.011 / ((1000 - 400) \times 10^{-6}) =$ **550 cfm**

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/CO2.html** in a browser.

⌂ All materials

F. WRITE IN EXAM Model answer

Write this in the exam

Assume (class notes): N_p = 0.011 cfm CO_2/person, C_{in} = 1000 ppm; Khulna 400 ppm, Dhaka 800 ppm.

CO_2 balance: $V = N\,N_p / (C_{in} - C_{out})$

Khulna: $V = 30 \times 0.011 / [(1000 - 400) \times 10^{-6}]$ = **550 cfm = 260 L/s**

Dhaka: $V = 0.33 / [(1000 - 800) \times 10^{-6}]$ = **1650 cfm = 779 L/s** (3 times Khulna)

IAQ control: (1) source control (low-VOC materials, no smoking, local exhaust); (2) filtration (MERV 8–13 for dusty Dhaka air); (3) dilution ventilation with CO_2-sensor-controlled dampers; keep RH 40–60%.

G. UNDERSTAND Common mistakes

For understanding – do not write

- **ppm to fraction.** 1 ppm = 10^{-6}.
- **cfm to L/s:** × 0.4719.
- **State the assumed N_p and C_{in}**: they are not on the paper.

10. Room cooling load with windows on two walls: 2024 7(c)

Topic B8 · 15 marks ·

A. Problem

Two of the wall faces, N and W, contain double-glazed windows (see figure). There is no heat gain through other walls. Find the required cooling capacity. Outside 43 °C DBT, 24 °C WBT; U of wall, roof, floor 1.78, 1.3, 1.2 W/m²K; ETD wall 25 °C, roof 30 °C; SHG 300 W/m², SC 0.86; lights 33 W/m² floor; 4 occupants (90 W sensible + 40 W latent each); appliances 600 W sensible + 400 W latent; infiltration 0.5 ACH. (Inside condition printed as "43 °C, 55%": misprint, taken as 25 °C / 50%.)

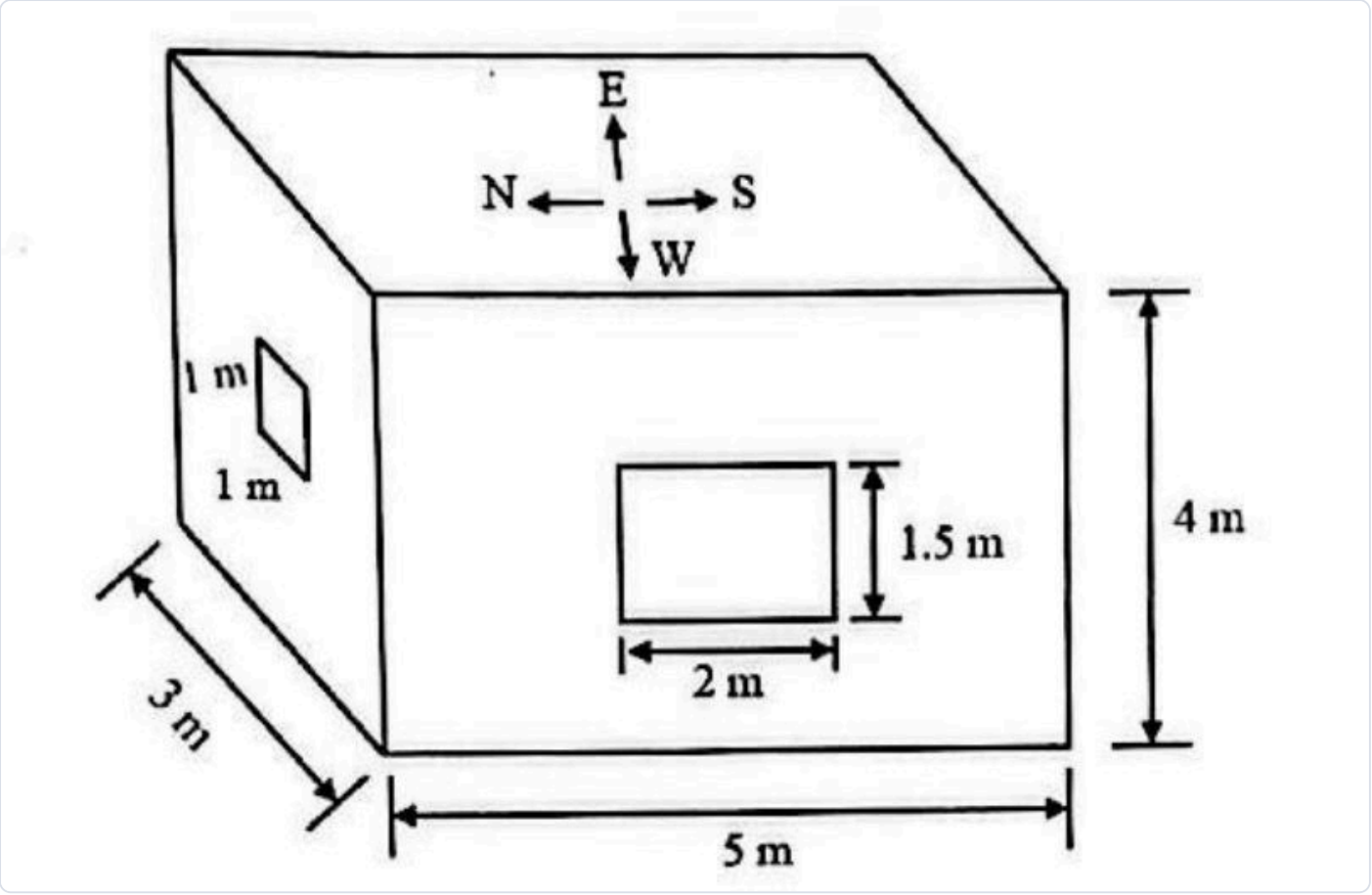


Figure given with the question

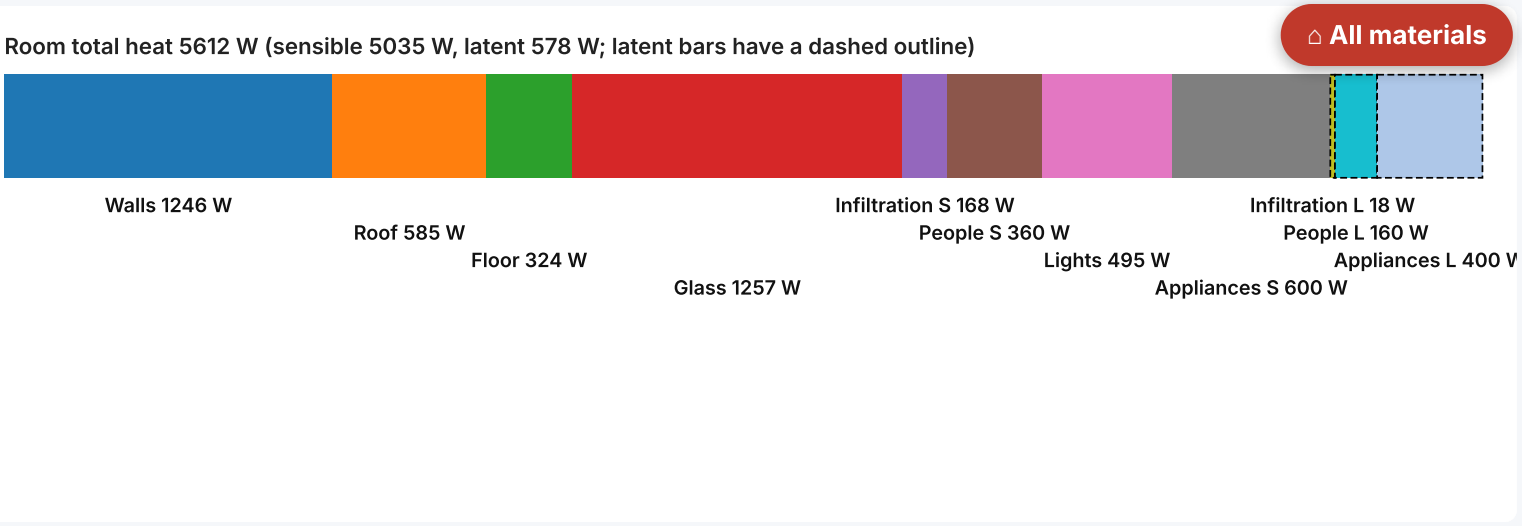
B. UNDERSTAND How the cycle/process works

For understanding – do not write

Cooling load = sum of every heat gain into the room. Two families: **sensible** (raises temperature) and **latent** (adds moisture). Each line uses one simple formula: walls and roof U A ETD (ETD = equivalent temperature difference, includes sun on the wall), glass A(U ΔT + SHG × SC), infiltration m c_p ΔT and m h_{fg} ΔW, people and appliances as given, lights W/m² × floor area.

The interactive bar shows which items dominate, so you know where a mistake costs most.

C. UNDERSTAND Interactive graph



Biggest slices are the ones worth checking twice in the exam (walls and glass here).

RSH	**5035**	W	RLH	**578**	W	RTH	**5612**	W
RSHF	**0.90**		× safety factor	**7015**	W	Capacity	**1.99**	TR

Working (formula = substitution = answer)

RTH = RSH + RLH = 5035 + 578 = **5612 W**

RSHF = RSH / RTH = 5035 / 5612 = **0.90**

Design load = RTH × SF = 5612 × 1.25 = **7015 W** = 7015 / 3517 = **1.99 TR**

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/LOAD.html** in a browser.

F. WRITE IN EXAM Model answer

Assume: inside 25 °C / 50% (misprint), glass U = 3.12 W/m²K, floor ΔT = 18 K, CLF = 1, ρ = 1.095 kg/m³, W_i = 0.00992, W_o = 0.0107.

Areas: W wall 5 × 4 − 3 = 17 m²; N wall 3 × 4 − 1 = 11 m²; roof = floor = 15 m²; glass 4 m²; volume 60 m³.

Item	Calculation	W
Walls	1.78 × 28 × 25	1246
Roof	1.3 × 15 × 30	585
Floor	1.2 × 15 × 18	324
Glass	4[3.12 × 18 + 300 × 0.86]	1257
Infiltration (S)	m = 1.095 × 0.5 × 60/3600 = 0.00913 kg/s; × 1021.6 × 18	168
People (S)	4 × 90	360
Lights	33 × 15	495
Appliances (S)	given	600
RSH		**5035**
Infiltration (L)	0.00913 × 2501 × 10^3 × (0.0107 − 0.00992)	18
People (L)	4 × 40	160
Appliances (L)	given	400
RLH		**578**

RTH = 5035 + 578 = 5612 W; with safety factor 1.25: **7015 W ≈ 2.0 TR**

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Deducting the window from the wall area.** Wall area is net of glass.
- **Using ETD for glass.** Glass uses UΔT + SHG × SC.
- **State every assumption** in one line at the top.

11. Duct sizing by equal friction: 2020 6(c)

Topic B9 · 12 marks ·

A. Problem

Determine the dimensions of AB, BC, CD (figure) using the equal friction method. Choose a friction rate of 0.09 mm H_2O/m. Use $\Delta p_f/L = 0.002268\ Q^{1.852}/D^{4.973}$ mm H_2O/m (Q in m^3/s, D in m).

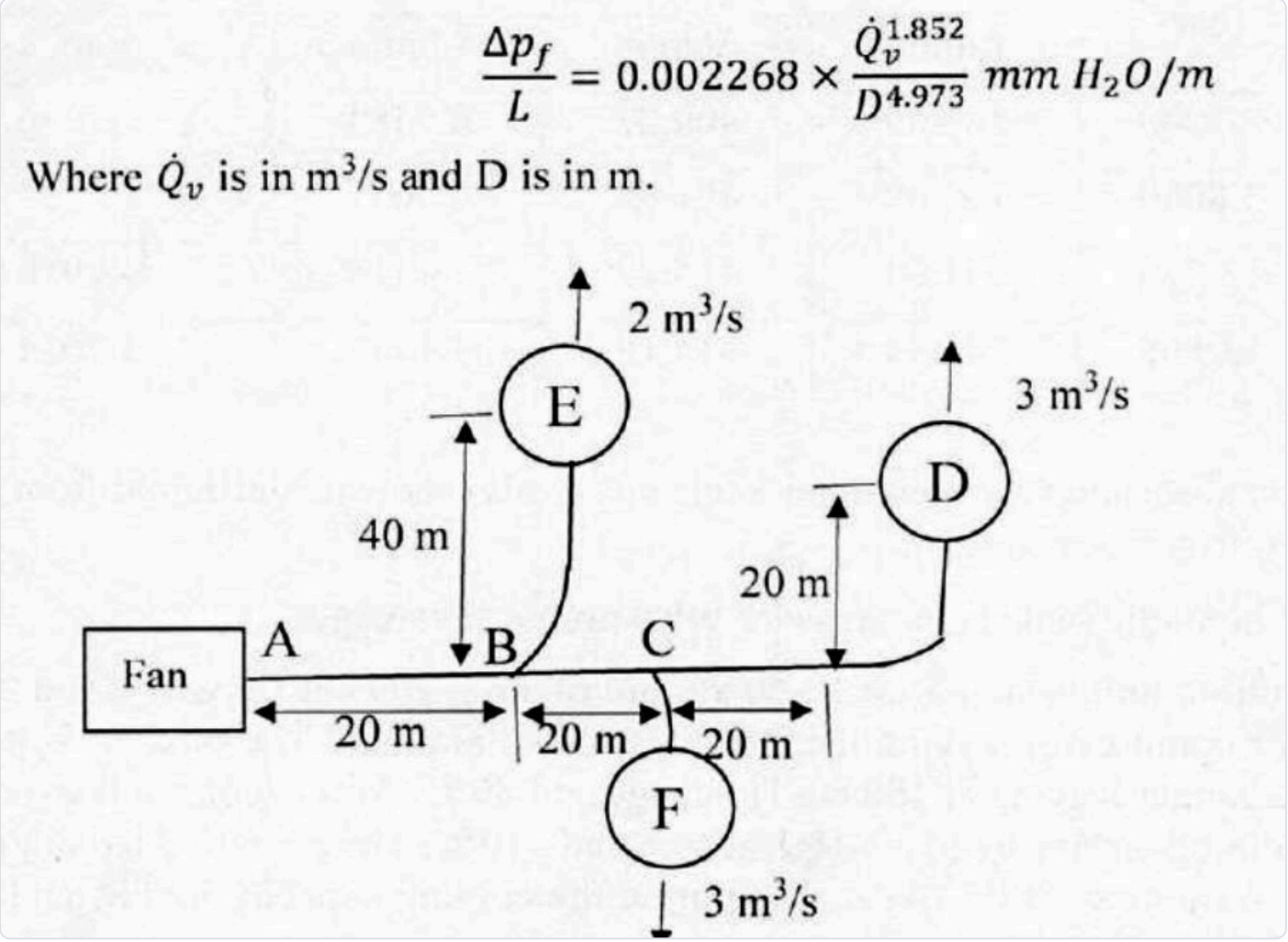


Figure given with the question

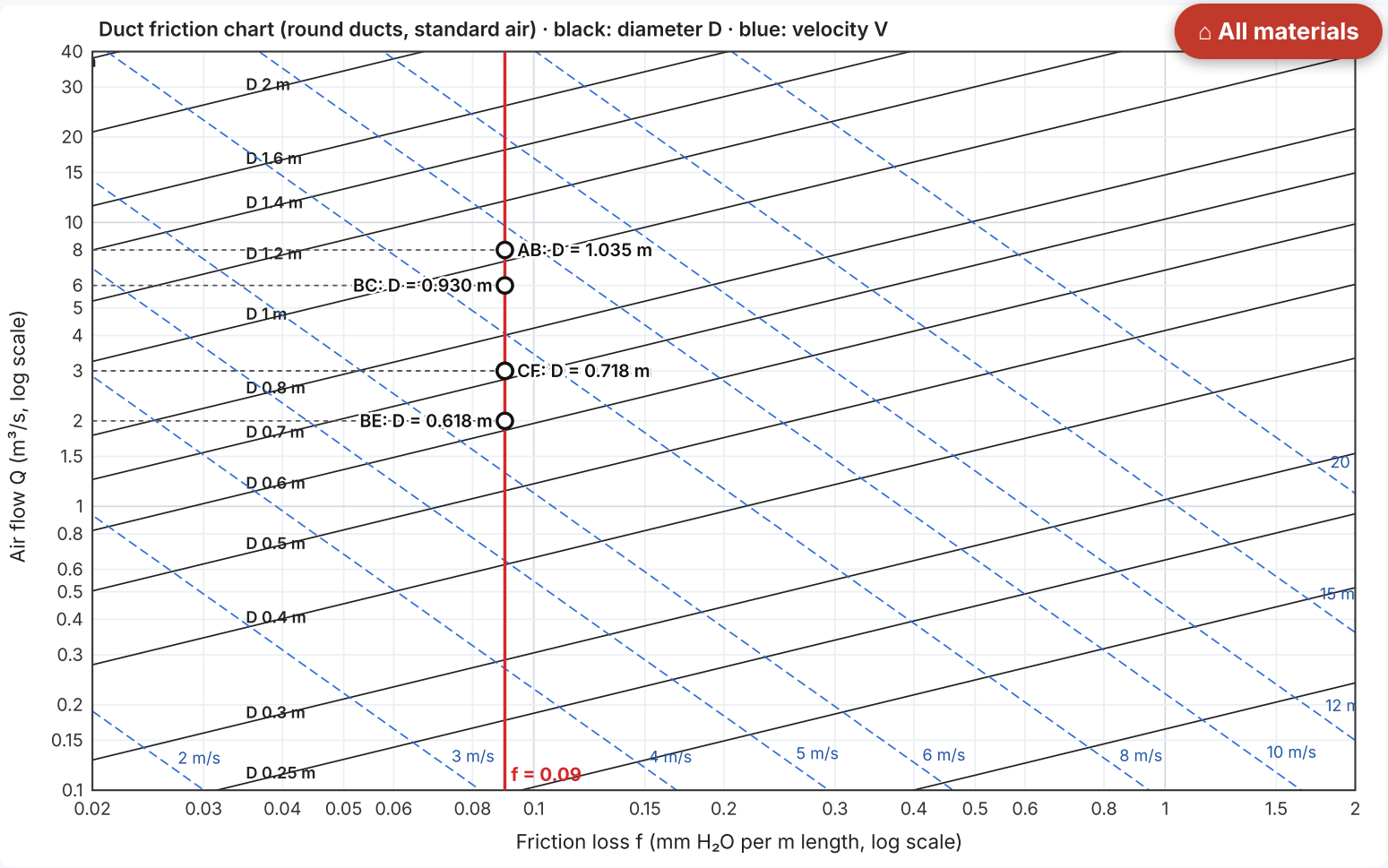
B. UNDERSTAND How the cycle/process works

For understanding – do not write

Equal friction method: every section is sized for the same pressure loss per metre, f. So on the friction chart every section sits on **one vertical line** (f), at its own airflow Q; read the diameter there.

Step 1 is always the flows: start at the far outlets and add backwards toward the fan. **Step 2** D from the chart or formula. **Step 3** (if rectangular) b = 0.548 D, a = 3b for 3:1. **Step 4** total friction = f × length of the longest run.

C. UNDERSTAND Interactive graph



Each section sits on the vertical friction-rate line; read its diameter from the black lines and its velocity from the blue lines. Hover to read any point.

Section	Q m³/s	L m	D (round) m	V m/s	p_v mm	f·L mm
AB	8	20	**1.035**	9.51	5.53	1.80
BC	6	20	**0.930**	8.84	4.78	1.80
CD	3	40	**0.718**	7.41	3.35	3.60
BE	2	40	**0.618**	6.68	2.73	3.60
CF	3		**0.718**	7.41	3.35	

Highlighted rows = index (longest) run, ΣL = 80 m → friction = 7.20 mm H_2O.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/DUCT-2020.html** in a browser.

F. WRITE IN EXAM Model answer

Method: equal friction, f = 0.09 mm H_2O/m, D from f = 0.002268 $Q^{1.852}/D^{4.973}$.

Flows (add outlets downstream): AB = 8, BC = 6, CD = 3, BE = 2, CF = 3 m^3/s

Sample: $D_{AB} = (0.002268 \times 8^{1.852} / 0.09)^{1/4.973} = 1.035$ m

$V = Q/(\pi D^2/4) = 8/(\pi \times 1.035^2/4) = 9.51$ m/s

Section	Q (m^3/s)	D = (0.002268 $Q^{1.852}/f)^{1/4.973}$ (m)	V = $Q/(\pi D^2/4)$ (m/s)
AB	8	1.035	9.51
BC	6	0.930	8.84
CD	3	0.718	7.41
BE	2	0.618	6.68
CF	3	0.718	7.41

Index (longest) run = 20 + 20 + 40 = 80 m → friction loss = 80 × 0.09 = **7.20 mm H_2O**

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Wrong flows.** A section carries the sum of all outlets downstream of it, not the outlet next to it.
- **Rectangle from area.** Do not set ab = $\pi D^2/4$; use the equivalent-diameter formula (same friction, not same area).
- **Units.** f in mm H_2O per m, Q in m^3/s, D in m.

12. Duct sizing by equal friction: 2021 8(d)

Topic B9 · 14 marks ·

A. Problem

For the duct system shown, determine the dimensions of AB, BC, CD and DE of rectangular ducts with aspect ratio 3:1, assuming a friction rate of 0.1 mm H_2O per m length.

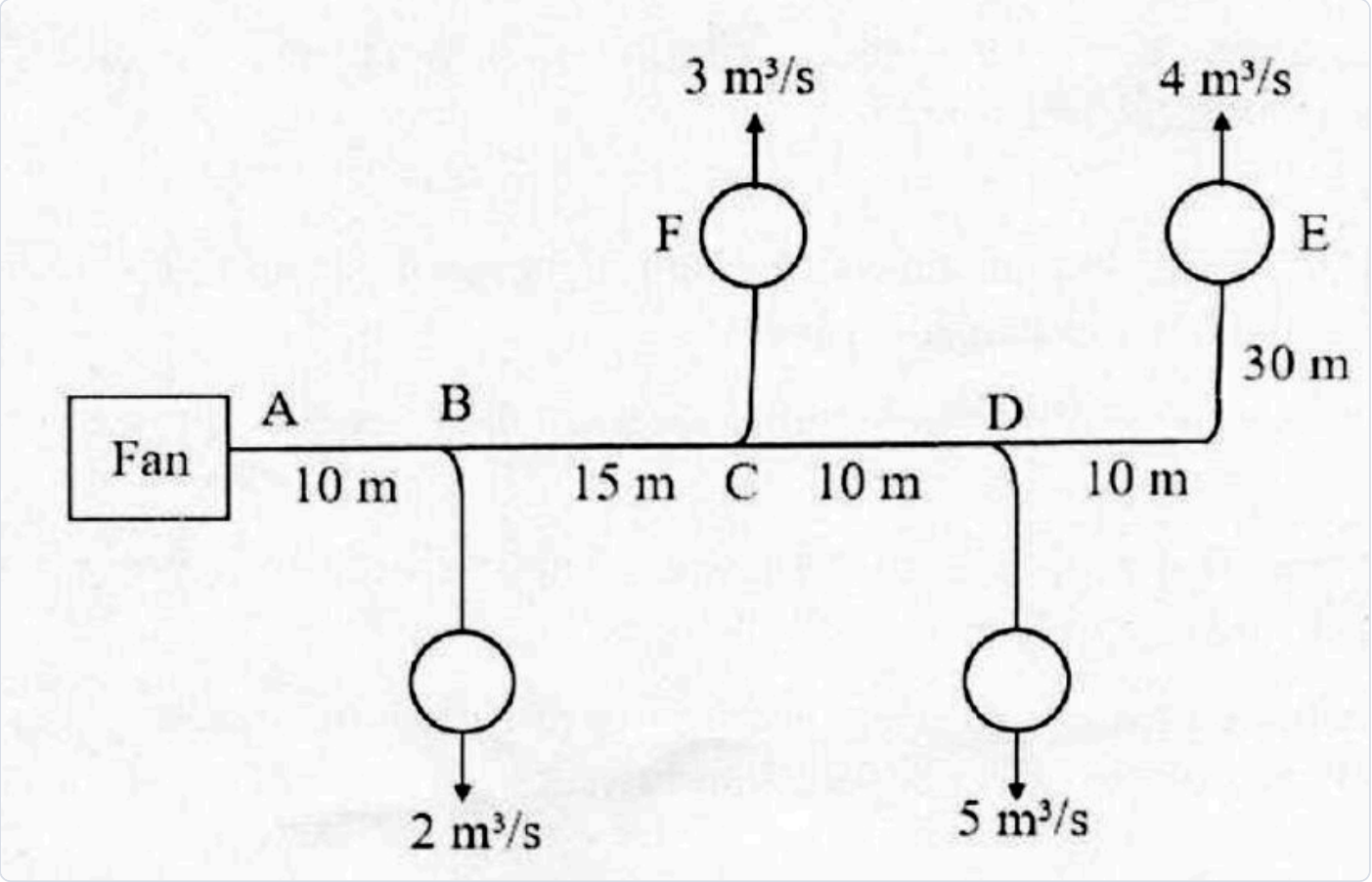


Figure given with the question

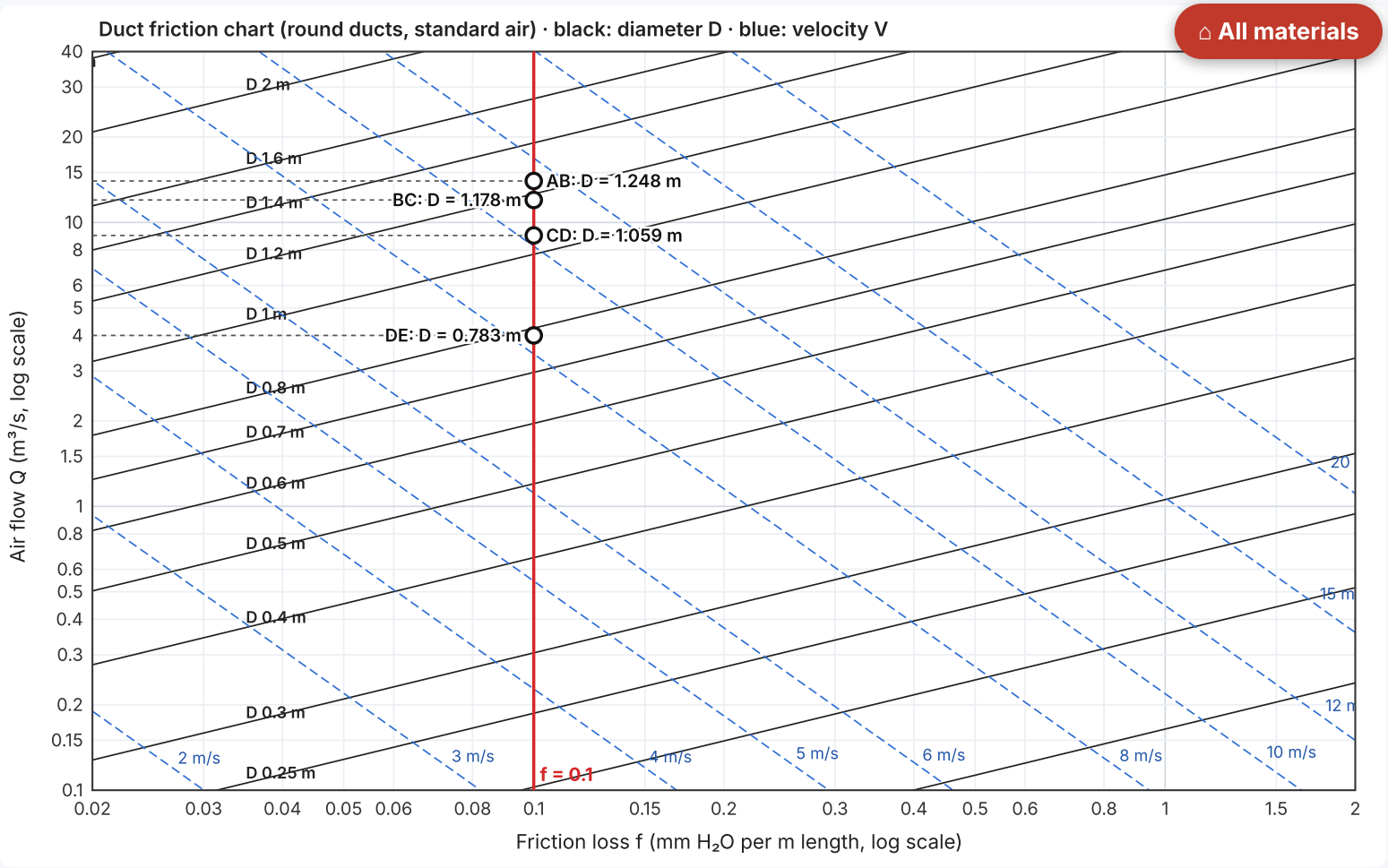
B. UNDERSTAND How the cycle/process works

For understanding – do not write

Equal friction method: every section is sized for the same pressure loss per metre, f. So on the friction chart every section sits on **one vertical line** (f), at its own airflow Q; read the diameter there.

Step 1 is always the flows: start at the far outlets and add backwards toward the fan. **Step 2** D from the chart or formula. **Step 3** (if rectangular) b = 0.548 D, a = 3b for 3:1. **Step 4** total friction = f × length of the longest run.

C. UNDERSTAND Interactive graph



Each section sits on the vertical friction-rate line; read its diameter from the black lines and its velocity from the blue lines. Hover to read any point.

Section	Q m³/s	L m	D (round) m	b = 0.547D	a = 3b	V m/s	p_v mm	f·L mm
AB	14	10	**1.248**	0.683	**2.050**	10.00	6.11	1.00
BC	12	15	**1.178**	0.645	**1.935**	9.61	5.65	1.50
CD	9	10	**1.059**	0.580	**1.739**	8.93	4.88	1.00
DE	4	40	**0.783**	0.428	**1.285**	7.26	3.23	4.00

Highlighted rows = index (longest) run, ΣL = 75 m → friction = 7.50 mm H_2O.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/DUCT-2021.html** in a browser.

F. WRITE IN EXAM Model answer

Write this in the exam

Method: equal friction, f = 0.1 mm H_2O/m, D from $f = 0.002268\ Q^{1.852}/D^{4.973}$, rectangular a:b = 3:1 with $D_e = 1.3(ab)^{0.625}/(a+b)^{0.25} \Rightarrow b = 0.548\ D_e$.

Flows (add outlets downstream): AB = 14, BC = 12, CD = 9, DE = 4 m^3/s

Sample: $D_{AB} = (0.002268 \times 14^{1.852} / 0.1)^{1/4.973} = 1.248$ m

b = 0.548 D = 0.548 × 1.248 = 0.683 m; a = 3b = 2.050 m; V = Q/(a b) = 14/(2.050 × 0.683) = 10.00 m/s

Section	Q (m^3/s)	D = $(0.002268\ Q^{1.852}/f)^{1/4.973}$ (m)	b = 0.548 D (m)	a = 3b (m)	V = Q/(a b) (m/s)
AB	14	1.248	0.683	**2.05 × 0.68**	10.00
BC	12	1.178	0.645	**1.94 × 0.65**	9.61
CD	9	1.059	0.580	**1.74 × 0.58**	8.93
DE	4	0.783	0.428	**1.29 × 0.43**	7.26

Index (longest) run = 10 + 15 + 10 + 40 = 75 m → friction loss = 75 × 0.1 = **7.50 mm H_2O**

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Wrong flows.** A section carries the sum of all outlets downstream of it, not the outlet next to it.
- **Rectangle from area.** Do not set ab = $\pi D^2/4$; use the equivalent-diameter formula (same friction, not same area).
- **Units.** f in mm H_2O per m, Q in m^3/s, D in m.

13. Duct sizing by equal friction: 2023 8(d)

Topic B9 · 15 marks ·

A. Problem

Determine the size of the duct system shown in the figure using the equal friction method, assuming a pressure drop of 0.1 mm H_2O/m length.

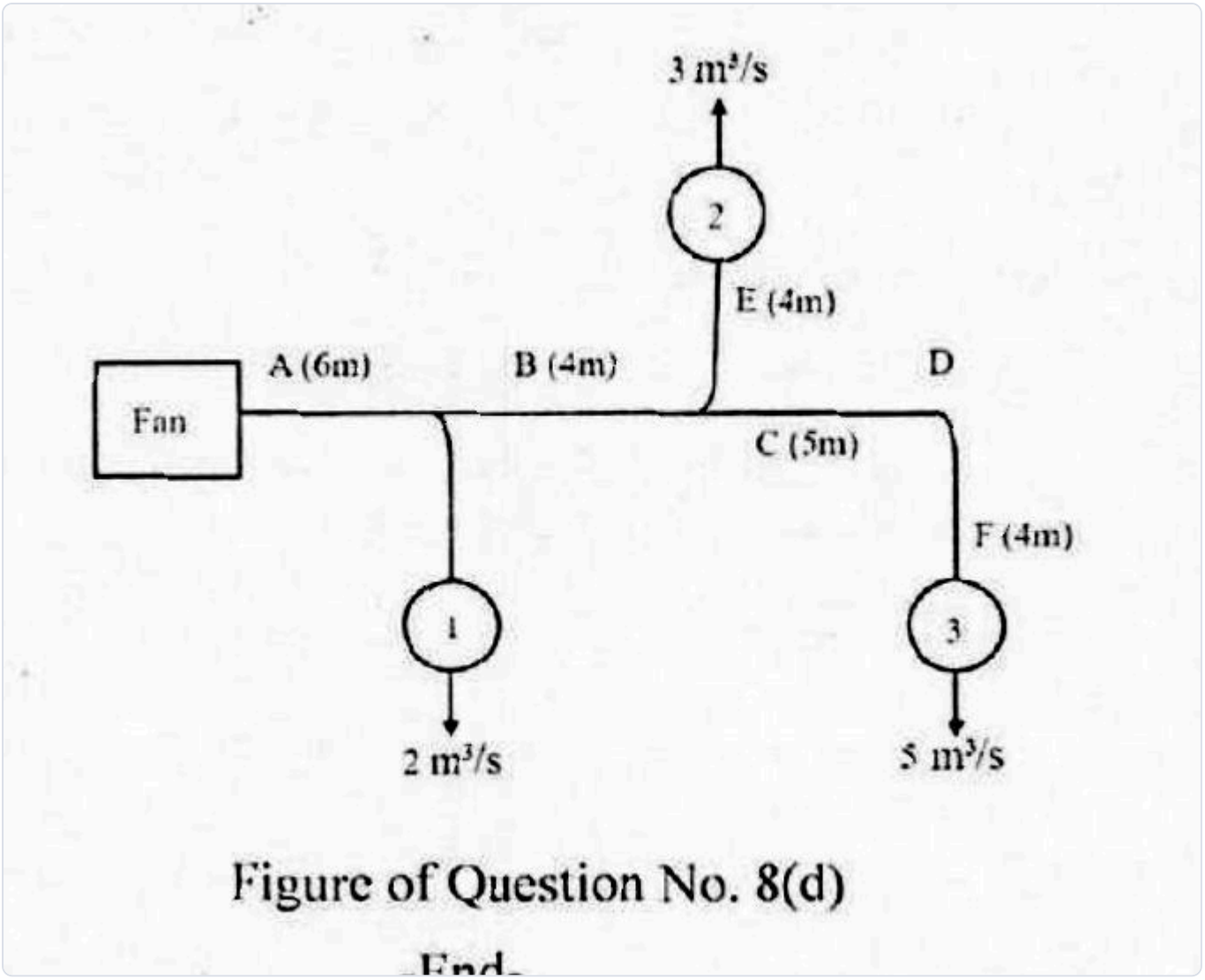


Figure of Question No. 8(d)

End

Figure given with the question

B. UNDERSTAND How the cycle/process works

For understanding – do not write

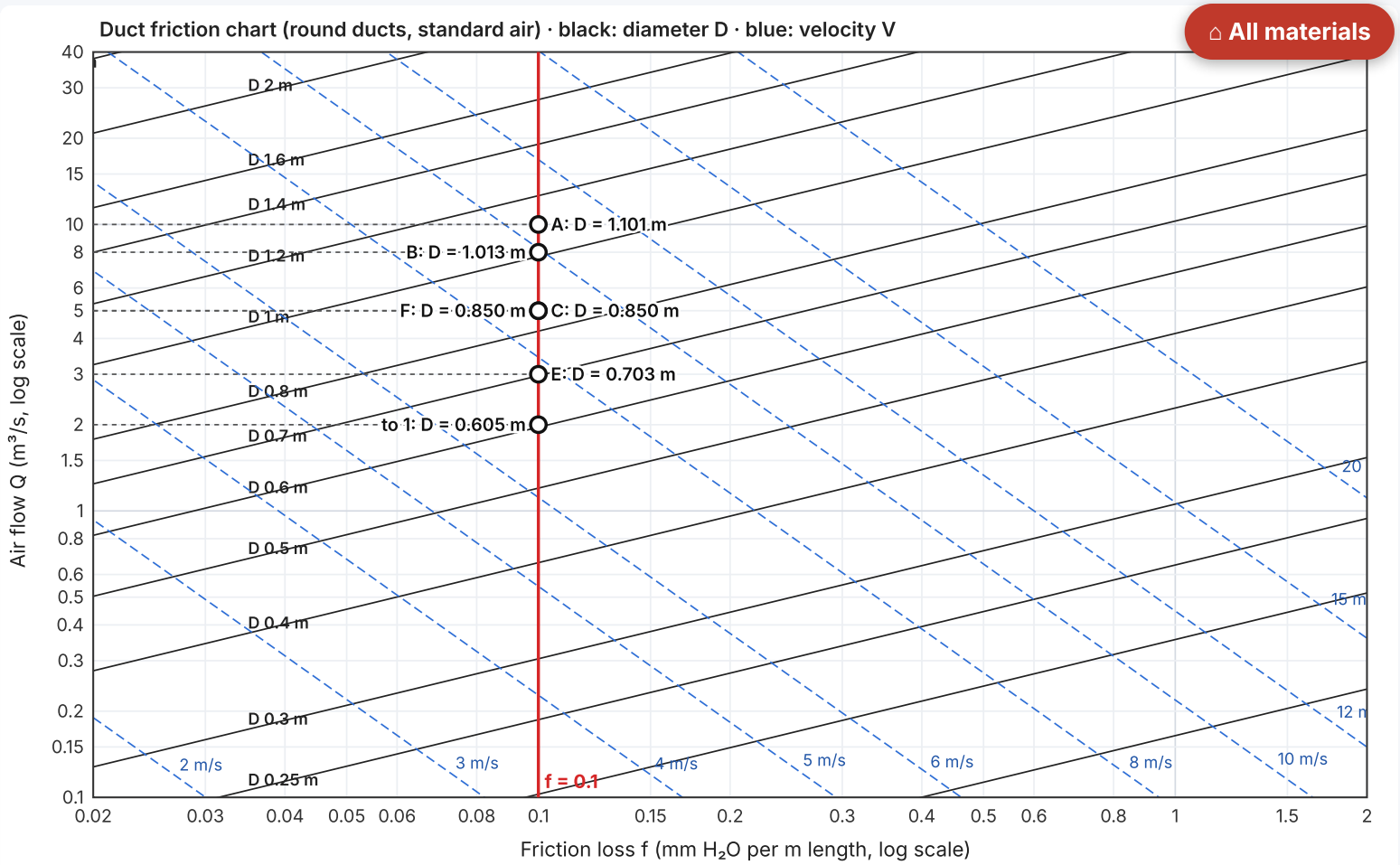
⌂ All materials

Equal friction method: every section is sized for the same pressure loss per metre, f. So on the friction chart every section sits on **one vertical line** (f), at its own airflow Q; read the diameter there.

Step 1 is always the flows: start at the far outlets and add backwards toward the fan. **Step 2** D from the chart or formula. **Step 3** (if rectangular) b = 0.548 D, a = 3b for 3:1. **Step 4** total friction = f × length of the longest run.

No aspect ratio is given, so round ducts are the answer; the rectangular 3:1 sizes are an optional extra (tick “rectangular” below).

C. UNDERSTAND Interactive graph



Each section sits on the vertical friction-rate line; read its diameter from the black lines and its velocity from the blue lines. Hover to read any point.

Section	Q m³/s	L m	D (round) m	V m/s	p_V mm	f·L mm
A	10	6	**1.101**	10.51	6.75	0.60
B	8	4	**1.013**	9.92	6.02	0.40
C	5	5	**0.850**	8.80	4.74	0.50
F	5	4	**0.850**	8.80	4.74	0.40
E	3	4	**0.703**	7.73	3.65	0.40
to 1	2		**0.605**	6.97	2.97	

Highlighted rows = index (longest) run, ΣL = 19 m → friction = 1.90 mm H_2O.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/DUCT-2023.html** in a browser.

F. WRITE IN EXAM Model answer

Write this in the exam

Method: equal friction, f = 0.1 mm H_2O/m, D from $f = 0.002268\ Q^{1.852}/D^{4.973}$.

Flows (add outlets downstream): A = 10, B = 8, C = 5, F = 5, E = 3, to 1 = 2 m^3/s

Sample: $D_A = (0.002268 \times 10^{1.852} / 0.1)^{1/4.973} = 1.101$ m

$V = Q/(\pi D^2/4) = 10/(\pi \times 1.101^2/4) = 10.51$ m/s

Section	Q (m^3/s)	D = $(0.002268\ Q^{1.852}/f)^{1/4.973}$ (m)	V = $Q/(\pi D^2/4)$ (m/s)
A	10	1.101	10.51
B	8	1.013	9.92
C	5	0.850	8.80
F	5	0.850	8.80
E	3	0.703	7.73
to 1	2	0.605	6.97

Index (longest) run = 6 + 4 + 5 + 4 = 19 m → friction loss = 19 × 0.1 = **1.90 mm H_2O**

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Wrong flows.** A section carries the sum of all outlets downstream of it, not the outlet next to it.
- **Rectangle from area.** Do not set ab = $\pi D^2/4$; use the equivalent-diameter formula (same friction, not same area).
- **Units.** f in mm H_2O per m, Q in m^3/s, D in m.

14. Duct sizing by equal friction: 2024 8(c)

Topic B9 · 15 marks ·

A. Problem

For the duct system shown in the figure, determine the dimensions of rectangular ducts with aspect ratio 3:1. Assume all dynamic loss coefficients are 0.8. *(Friction rate not printed: assume 0.1 mm H_2O/m and state it.)*

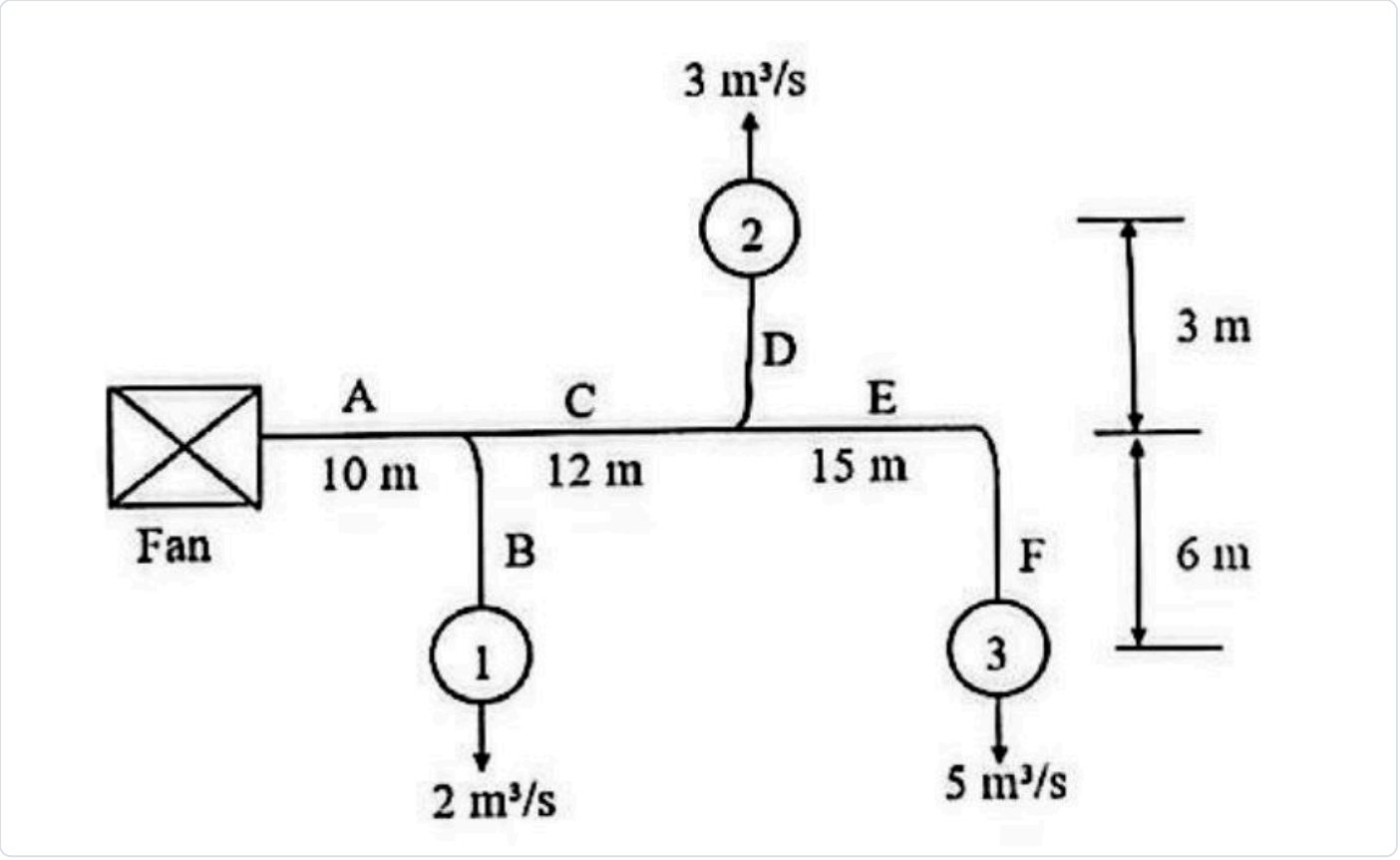


Figure given with the question

B. UNDERSTAND How the cycle/process works

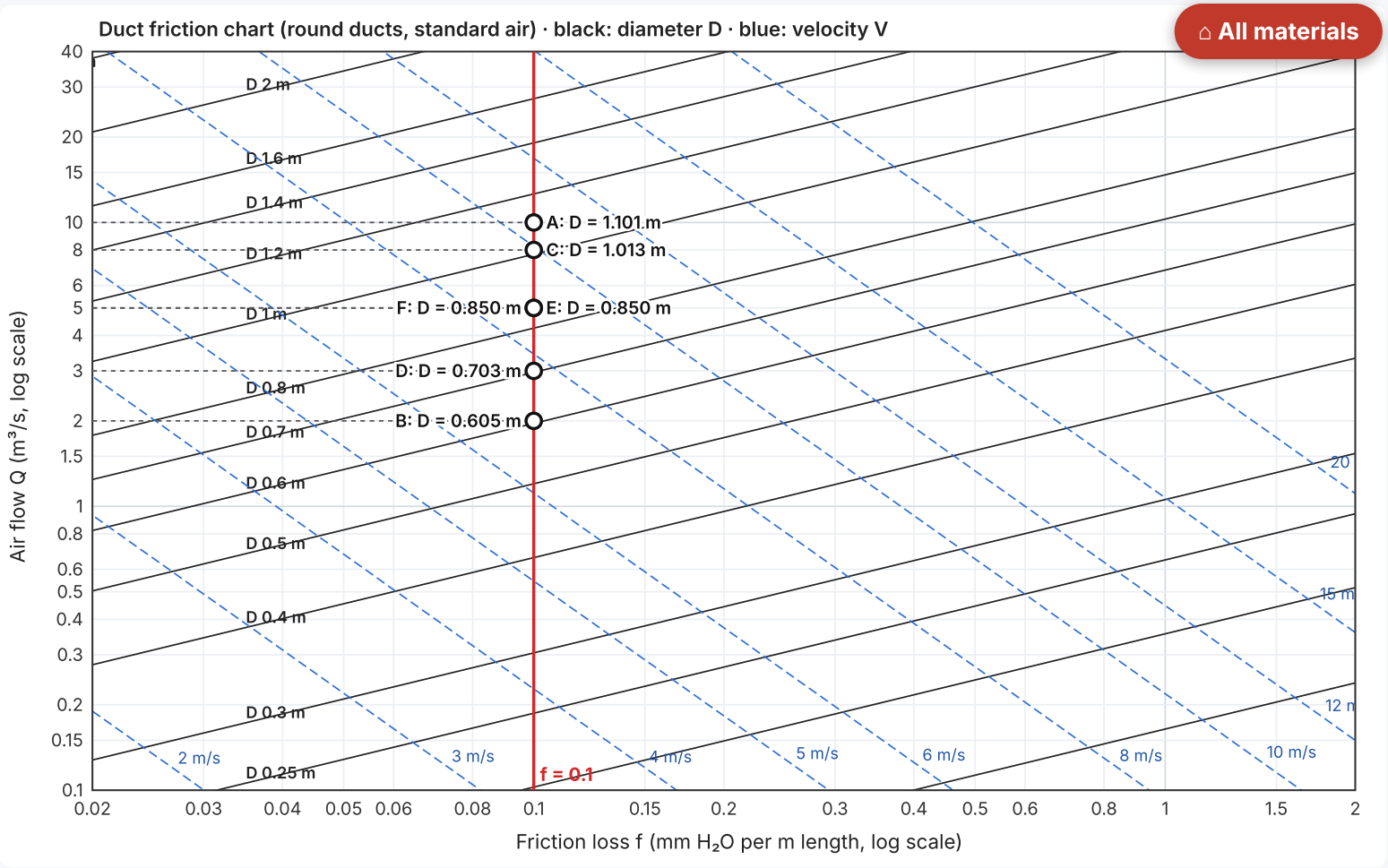
For understanding – do not write

Equal friction method: every section is sized for the same pressure loss per metre, f. So on the friction chart every section sits on **one vertical line** (f), at its own airflow Q; read the diameter there.

Step 1 is always the flows: start at the far outlets and add backwards toward the fan. **Step 2** D from the chart or formula. **Step 3** (if rectangular) b = 0.548 D, a = 3b for 3:1. **Step 4** total friction = f × length of the longest run.

Dynamic losses (bends, junctions) = K × velocity pressure $p_v = K\,\rho V^2/2$; in mm H_2O, $p_v = 1.2V^2/(2 \times 9.81)$. Add them along the index run to the friction loss to get fan pressure.

C. UNDERSTAND Interactive graph



Each section sits on the vertical friction-rate line; read its diameter from the black lines and its velocity from the blue lines. Hover to read any point.

Section	Q m³/s	L m	D (round) m	b = 0.547D	a = 3b	V m/s	p_v mm	f·L mm
A	10	10	**1.101**	0.603	**1.808**	9.18	5.15	1.00
B	2	6	**0.605**	0.331	**0.993**	6.09	2.26	0.60
C	8	12	**1.013**	0.555	**1.664**	8.67	4.59	1.20
D	3	3	**0.703**	0.385	**1.155**	6.75	2.79	0.30
E	5	15	**0.850**	0.466	**1.397**	7.69	3.61	1.50
F	5	6	**0.850**	0.466	**1.397**	7.69	3.61	0.60

Highlighted rows = index (longest) run, ΣL = 43 m → friction = 4.30 mm H_2O; dynamic losses = 9.46 mm; fan pressure ≈ **13.76 mm H_2O**.

Printed snapshot at the exam values. Interactive version (sliders, step-through, hover read-out): open **problems/DUCT-2024.html** in a browser.

F. WRITE IN EXAM Model answer

Method: equal friction, f = 0.1 mm H_2O/m, D from $f = 0.002268\ Q^{1.852}/D^{4.973}$, rectangular a:b = 3:1 with $D_e = 1.3(ab)^{0.625}/(a+b)^{0.25} \Rightarrow b = 0.548\ D_e$.

Flows (add outlets downstream): A = 10, B = 2, C = 8, D = 3, E = 5, F = 5 m³/s

Sample: $D_A = (0.002268 \times 10^{1.852} / 0.1)^{1/4.973} = 1.101$ m

b = 0.548 D = 0.548 × 1.101 = 0.603 m; a = 3b = 1.808 m; V = Q/(a b) = 10/(1.808 × 0.603) = 9.18 m/s

Section	Q (m³/s)	D = (0.002268 $Q^{1.852}/f)^{1/4.973}$ (m)	b = 0.548 D (m)	a = 3b (m)	V = Q/(a b) (m/s)
A	10	1.101	0.603	**1.81 × 0.60**	9.18
B	2	0.605	0.331	**0.99 × 0.33**	6.09
C	8	1.013	0.555	**1.66 × 0.55**	8.67
D	3	0.703	0.385	**1.15 × 0.38**	6.75
E	5	0.850	0.466	**1.40 × 0.47**	7.69
F	5	0.850	0.466	**1.40 × 0.47**	7.69

Index (longest) run = 10 + 12 + 15 + 6 = 43 m → friction loss = 43 × 0.1 = **4.30 mm H_2O**

Velocity pressure $p_v = \rho V^2/2 = 1.2V^2/(2 \times 9.81)$ mm H_2O: C: $1.2 \times 8.67^2/19.62 = 4.59$; E: $1.2 \times 7.69^2/19.62 = 3.61$; F: $1.2 \times 7.69^2/19.62 = 3.61$

Dynamic loss = 0.8 × (4.59 + 3.61 + 3.61) = 9.46 mm H_2O

Fan pressure ≈ 4.30 + 9.46 = **13.8 mm H_2O**

G. UNDERSTAND Common mistakes

For understanding – do not write

- **Wrong flows.** A section carries the sum of all outlets downstream of it, not the outlet next to it.
- **Rectangle from area.** Do not set ab = $\pi D^2/4$; use the equivalent-diameter formula (same friction, not same area).
- **Units.** f in mm H_2O per m, Q in m³/s, D in m.

Chart library (the charts at the back of the book, interactive)

Open these on their own to practise reading charts. All work offline.

- R22 P–h chart · R134a P–h chart · Ammonia P–h chart (switch enthalpy zero between chart and table basis)
- Psychrometric chart (sea level)
- Duct friction chart

Abbreviations

VCR	vapour compression refrigeration
COP	coefficient of performance
RE	refrigerating effect ($h_1 - h_4$)
TR	ton of refrigeration = 210 kJ/min ≈ 3.5 kW
IIR	International Institute of Refrigeration (chart enthalpy zero: h = 200 at 0 °C liquid)
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers (table zero: h = 0 at −40 °C liquid)
DBT / WBT / DPT	dry bulb / wet bulb / dew point temperature
RH	relative humidity
W	humidity ratio, kg water per kg dry air
ADP	apparatus dew point (coil surface temperature)
X / BPF	bypass factor
RSH / RLH / RTH	room sensible / latent / total heat
OASH / OALH / OATH	outside-air sensible / latent / total heat
GTH	grand total heat
RSHF	room sensible heat factor
ETD	equivalent temperature difference
SHG	solar heat gain
SC	shading coefficient
ACH	air changes per hour
cmm / cfm	cubic metres per minute / cubic feet per minute
IAQ	indoor air quality